MAP Open Intrusion Interface

Resource Model

en Technical Description

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0 Definitions, Acronyms and Abbreviations

OII	Open Intrusion Interface
REST	Representational State Transfer
HTTP	Hypertext Transfer Protocol
TCP	Transmission Control Protocol
TLS	Transport Layer Security
IP	Internet Protocol
MAP	Modular Alarm Platform
Etag	Entity Tag
URI	Uniform Resource Identifier
bool	JSON value of “true” or “false”
<EmptyString>	An empty JSON string i.e. ""

1 Scope

This document defines the resources and operations provided by the MAP5000 via the Open Intrusion Interface (OII). This specification reuses the basic communication mechanisms and design rules as defined in the OII Base Specification.

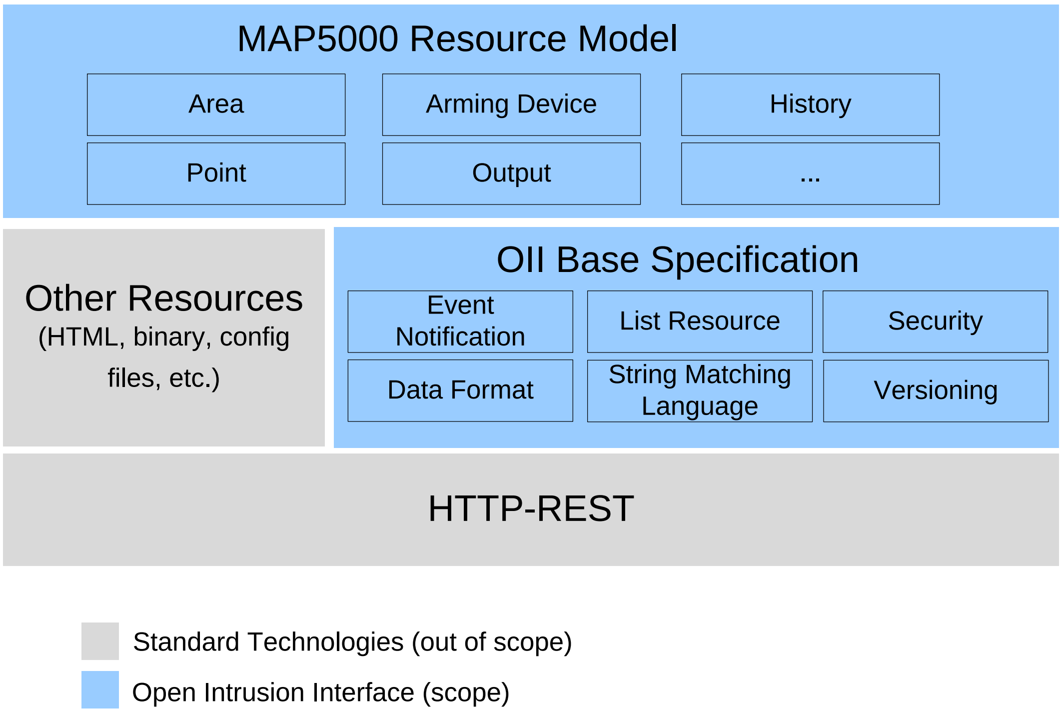

Fig. 1 OII Specification Overview

As the OII follows a RESTful approach, the herein defined resource model specifies a number of resources. For each resource, its data format (as a JSON object) is specified. Each resource supports simple information retrieval via a GET request, which will provide the JSON based resource representation.

Depending on the resource, more operations may be available. For these operations, the verb (i.e. PUT, POST, DELETE), the query parameters (e.g. url) and the potentially required request information is specified.

2 Resource Model Overview

In the table below an overview is provided for all resources in the

OII Resource Model version 1.1

Please note that the resource URLs are only valid for this version of the resource model. In future versions (even with minor version changes e.g. 1.1, 1.2) the resources may be located under different URLs. Thus, a client should use the description resource ("/desc") to discover the version of the resource model as well as the URLs to the major resources of the panel.

Resource Type	Resource URL	Description	Supported Operations
OII.desc.1	/desc	Descriptive information of the MAP5000 including versioning information.	Status Retrieval
IN.config.1	/config	Resource describing the configuration of the MAP5000 with regards to areas and devices.	Status Retrieval
IN.panel.1.1	/panel	Resource providing the current status of the panel	Status Retrieval Restart
OII.subList.1	/sub	Central resource to create subscription for any resource available in the MAP5000	Status Retrieval Subscription
OII.sub.1	/sub/*	Individual subscription. Concrete link is provided as part of the subscription process.	Status Retrieval Fetch Events Unsubscription
IN.incList.1	/inc	List of all incidents in the MAP.	Status Retrieval Handle
IN.inc.1.1	/inc/*	Individual incident.	Status Retrieval Handle
IN.time.1	/time	Current time of the panel.	Status Retrieval Set Time

Resource Type	Resource URL	Description	Supported Operations
IN.userList.1	/users	Activation/Deactivation of configured MAP users.	Status Retrieval De-/Activate User
IN.network.1	/network	Inspect and change network settings	Status Retrieval Activate DHCP Set Static IP Address
IN.history.1	/history	Access to the panel history data including ENGrade 3 Mandatory events	Status Retrieval
IN.ipcHistory.1	/ipchistory	Access to the IPC history data only	Status Retrieval
IN.areaList.1	/areas	List of all areas configured in the MAP5000.	Status Retrieval Arm/Disarm Start/Stop Walktest Start/Stop Motion Detector Test Start/Stop Chime mode
IN.area.1	/[Area SIID]	Individual area.	Status Retrieval Arm/Disarm Start/Stop Walktest Start/Stop Motion Detector Test Start/Stop Chime mode Start Bell Test
IN.internalProgramList.1	/internalprograms	List all internal programs configured	Status Retrieval Activation/Deactivation
IN.internalProgram.1	/internalprogram/*	Individual internal program (1 to 14)	Status Retrieval Activation/Deactivation IP Arming Info
IN.walktestList.1	/walktests	List of all currently active walktests in the system	Status Retrieval Stop Walktest
IN.walktest.1	/walktest/*	Individual walktest	Status Retrieval Stop Walktest
IN.deviceList.1	/devices	List of all devices configured in the panel	Status Retrieval
IN.supervConnList.1	/supervisedConns	List of Connections to other systems like BIS or IPC, OII.	Status Retrieval Disable/Enable
IN.supervConn.1	/[SIID]	Individual supervised connection	Status Retrieval Disable/Enable

Resource Type	Resource URL	Description	Supported Operations
IN.oii.1	/1.1.system.15.1	OII interface resource. Only used for reporting of user code tamper on OII.	Status Retrieval Disable/Enable
IN.keypadList.1	/keypads	List of keypads in the system	Status Retrieval Disable/Enable Activate/Deactivate
IN.keypad.1	/[SIID]	Individual keypad	Status Retrieval Disable/Enable Activate/Deactivate Firmware Version
IN.lsnGwList.1	/lsnGateways	List of LSN gateways in the system	Status Retrieval Disable/Enable
IN.lsnGateway.1	/[SIID]	Individual gateway	Status Retrieval Disable/Enable Firmware Version
IN.powerSupList.1	/powerSupplies	List of MAP power supplies	Status Retrieval Disable/Enable
IN.powerSupply.1	/[SIID]	Individual powersupply	Status Retrieval Disable/Enable Firmware Version
IN.deModule.1	/[DEModule SIID]	DE Module of the MAP	Status Retrieval Disable/Enable Firmware Version
IN.blocklockList.1	/blocklocks	List of blocklocks	Status Retrieval Disable/Enable
IN.blocklock.1	/[SIID]	Individual blocklock device	Status Retrieval Disable/Enable Bypass/Unbypass
IN.smartkeyList.1	/smartkeys	List of smartkeys	Status Retrieval Disable/Enable
IN.smartkey.1	/[SIID]	Individual smartkey device	Status Retrieval Disable/Enable Bypass/Unbypass
IN.outputList.1	/outputs	List of outputs configured with the system including digital outputs, sirens, etc.	Status Retrieval Disable/Enable Turn On/Off
IN.output.1	/[SIID]	Individual output	Status Retrieval Disable/Enable Turn On/Off

Resource Type	Resource URL	Description	Supported Operations
IN.pointList.1	/points	List of points in the system. Points represent detectors like PIRs, glass break detectors or end of line resistors	Status Retrieval Disable/Enable
IN.point.1	/[SIID]	Individual point	Status Retrieval Disable/Enable Bypass/Unbypass
IN.fireDetList.1	/fireDetectors	List of fire detectors in the system	Status Retrieval Disable/Enable
IN.fireDetector.1	/[SIID	Individual fire detector	Status Retrieval Disable/Enable Bypass/Unbypass
IN.printer.1	/[Printer SIID]	Printer of the MAP	Status Retrieval Disable/Enable
IN.communicato r.1	/[AT2000/3000 SIID]	AT2000 or AT3000 configured for the MAP	Status Retrieval Disable/Enable
IN.keyswitchList .1	/keyswitches	List of keyswitches in the system	Status Retrieval Disable/Enable
IN.keyswitch.1	/[SIID]	Individual keyswitch	Status Retrieval Disable/Enable
IN.couplerList.1	/couplers	List of couplers in the system	Status Retrieval Disable/Enable
IN.coupler.1	/[SIID]	Individual coupler	Status Retrieval Disable/Enable
IN.lsnBusList.1	/lsnbuses	List of lsn buses (loops and stubs) in the system	Status Retrieval Disable/Enable
IN.lsnBus.1	/[SIID]	Individual lsn bus	Status Retrieval Disable/Enable
IN.lsnAuxList.1	/lsnauxs	List of LSN auxiliary power outlets in the system	Status Retrieval Disable/Enable
IN.lsnAux.1	/[SIID]	Individual LSN aux	Status Retrieval Disable/Enable
IN.batChgList.1	/batterychargers	List of battery chargers in the system	Status Retrieval Disable/Enable
IN.batterCharge r.1	/[SIID]	Individual battery charger	Status Retrieval Disable/Enable Bypass/Unbypass

Resource Type	Resource URL	Description	Supported Operations
IN.batteryList.1	/batteries	List of batteries in the system	Status Retrieval Disable/Enable
IN.battery.1	/[SIID]	Individual battery	Status Retrieval Disable/Enable Bypass/Unbypass
IN.mainsList.1	/mains	List of resources representing the mains connection of a power supply in the system	Status Retrieval Disable/Enable
IN.mains.1	/[SIID]	Individual main resource	Status Retrieval Disable/Enable Bypass/Unbypass
IN.gndFltList.1	/groundfaults	List of resources representing the ground faults of a power supply in the system	Status Retrieval Disable/Enable
IN. groundFault.1	/[SIID]	Individual ground fault resource	Status Retrieval Disable/Enable Bypass/Unbypass
IN.psCanOpList.1	/psCanOpList	List of resources representing the power supply CAN outputs in the system	Status Retrieval
IN.psCanOp.1	/[SIID]	Individual power supply CAN output resource	Status Retrieval
IN.powerDevList.1	/powerDevices	List of non BDB power supplies in the system like NEV300	Status Retrieval Disable/Enable
IN.powerDevice.1	/[SIID]	Individual power device	Status Retrieval Disable/Enable

Tab. 1 Resources on MAP5000

3 Description

The description of the panel is provided under “/desc”. Its format is specified in the OII Base Specification.

Overview

OII Type	Supported Operations
OII.desc.1	Status Retrieval

Object Structure

As defined in the OII Base Specification. The field “mainResources” will contain the links to the following resources:

Resource Type	Resource URL
OII.desc.1	/desc
IN.config.1	/config
OII.subList.1	/sub
IN.incList.1	/inc
IN.time.1	/time
IN.network.1	/network
IN.userList.1	/users
IN.history.1	/history
IN.ipcHistory.1	/ipchistory
IN.areaList.1	/areas
IN.deviceList.1	/devices
IN.internalProgramList.1	/internalprograms
IN.walktestList.1	/walktests
IN.supervConnList.1	/supervisedConns
IN.oii.1	/1.1.system.15.1
IN.keypadList.1	/keypads
IN.lsnGwList.1	/lsnGateways
IN.powerSupList.1	/powerSupplies
IN.deModule.1	/[DEModule SIID]
IN.blocklockList.1	/blocklocks
IN.smartkeyList.1	/smartkeys
IN.outputList.1	/outputs
IN.pointList.1	/points
IN.fireDetList.1	/fireDetectors
IN.printer.1	/[Printer SIID]
IN.communicator.1	/[AT2000/3000 SIID]
IN.keyswitchList.1	/keyswitches
IN.couplerList.1	/couplers
IN.lsnBusList.1	/lsnbuses

Resource Type	Resource URL
IN.lsnAuxList.1	/lsnauxs
IN.batChgList.1	/batterychargers
IN.batteryList.1	/batteries
IN.mainsList.1	/mains
IN.psCanOpList.1	/psCanOpList
IN.powerDevList.1	/powerDevices
IN.panel.1.1	/panel

4 Event Notification

The MAP5000 provides event notifications for all resources based on the event notification mechanism defined in the OII base specification. To this end, two types of resources are available.

- **Subscription Resource List: (/sub)** A central resource where a client can create a subscription. The location of the subscription resource is currently ("/sub").However, a client shall not assume the location to be fixed but rather take the location from the panel description resource, to anticipate changes in future interface versions. The subscription resource also provides an overview on all existing subscriptions.

- **Subscription Resource (/sub/*):** A resource representing individual, valid subscription of a client. This resource can be used to inspect the information about the current subscription, to fetch the events as well as to cancel the subscription. The link to the individual subscription is provided in the response to a subscription request. This resource is dynamically created and deleted during runtime. The panel assures that the subscription resource URL is unique even over power cycles of the panel. The URL shall be treated as an opaque identifier for the individual subscription. No semantics or sequence information shall be assumed by the client.

A client can decide which events to receive from the panel by setting the appropriate filters for the subscription. One client can also create multiple subscriptions so that the queue sizes and fetching behaviour fits to the expected occurrence of events. For example, it can be assumed that incident will only occur rarely while device state changes will happen often. Thus, a client may create two subscriptions; one subscription for incidents only and one subscription for devices only. Thus, incidents can be prioritized and fetched quicker as if they would be part of the same subscription.

A client that wants to subscribe to all resources of the panel may specify a filter with all elements in the following list. Please note that the URL of the resources may change over time. Thus, a client shall look up the URL for the given resource type from the description resource to be forward compatible.

Description	Current URL
All devices	/devices
All areas	/areas
All incidents	/inc/*
OII interface	/1.1.system.15.1
Supervised connections	/supervisedConns
Internal Programs	/internalPrograms
All users	/user/*
Panel resource	/panel
All walktests	/walktest/*

5 Configuration

A configuration file is provided that describes the areas, internal programs, devices and their relationship. A client can download the configuration on start up to understand which items are configured in the MAP5000 system. In particular, the relationship between areas and devices as well as among devices is given in this file. The relationship information is only given with the configuration. The resources themselves do not have links between each other, as the relationships cannot change during runtime.

Example

```
{
    "areaConfiguration": [
      {
        "name": "[Control Panel Area]",
        "siid": "1.1.ControlPanelArea.1.1",
        "disarmIfAllDisarmed": "NOCONSTRAINT",
        "autoDisarmed": false,
        "parentAreaType": "NOTAPARENT",
        "childAreaOptions": "NONE",
        "containedAreaList": [],
        "deviceList": [
          "1.1.Module.21001.001"
        ]
      }
    ],
  "internalProgramsConfiguration": {
    "name": "Internal program 01",
    "siid": "internalprogram/1",
    "deviceList": [
      "1.1.Point.21001.9"
    ]
  },
    "deviceConfiguration": [
      {
        "name": "Onboard I/O",
        "type": "POINT.ONBOARD",
        "partOfWalktest": true,
        "bypassable": true,
        "parentSIID": "",
        "siid": "1.1.Module.21001.001"
      }
    ]
}
```

Overview

OII Type	Supported Operations
IN.config.1	Status Retrieval

Object Structure

The configuration object has three major parts. In the first part of the configuration the list of all configured areas and their properties is provided. In addition, each area contains a list of all devices that are part of this area. The second part of the configuration is the internal program configuration which provides list of configured internal programs and the devices that are configured for these internal programs. The third part of the configuration is the device configuration. This part includes details about the devices.

Configuration object structure:

Name	Type	Value Range/Format	Description
@type	Array of strings	[IN.config.1]	Fixed type identifier
@self	String	URL starting with "/"	Link to the current resource
areaConfiguration	Array of area config. objects		An array of area configuration objects.
internalProgramsConfiguration	Array of internal program objects		An array of internal program configuration objects.
deviceConfiguration	Array of device config. objects		An array of device configuration objects.

Area configuration object structure:

Name	Type	Value Range/Format	Description
name	String	1 to 32 character long string.	The name used to identify the area.
siid	String	String in the form of A.B.<String>.C.D Where A, B, C and D are integer values String would be "ControlPanelArea" or "Area"	Auto generated unique Identifier.
disarmIfAllDisarmed	String	NOCONSTRAINT YES	The control panel area (CPA) can only be disarmed if all other system areas have been disarmed. (Applicable only for Control Panel Area)
autoDisarmed	Bool	true/false	When all the areas in the system are disarmed, the Control Panel Area (CPA), if armed, will disarm. (Applicable only for Control Panel Area)
parentAreaType	String	String identifier	Defines the Parent Area's arming relationship with child areas. NOTAPARENT : Area contains no child devices MASTER : Area contains child area and its relationship is explained by "ChildAreaOptions" attribute. SHARED: Area arms automatically when all child areas are armed. Area disarms automatically when one of the child area is disarmed

Name	Type	Value Range/Format	Description
			BANK: Can be armed only after all child areas are armed. PASSTHRU: Area arms automatically when one of the child areas is armed. Area disarms automatically when all child areas are disarmed.
childAreaOptions	String	String identifier	Configurable only for Master Area. NONE : Not arming/disarming relationship with its child areas ARMABLEIFCHILDARMED : Can be armed only if all child areas armed. ARMDISARMCHILDAREAS: Area will arm/disarm child areas when it is being armed/disarmed
containedAreaList	Array of Strings	Array of siids	List of all child area(s) SIIDs.
deviceList	Array of String	Array of strings in the form of **A.B.<String>.C.D** Where A, B, C and D are integer values	List of SIIDs of all devices configured in the area.

Internal Program configuration object structure:

Name	Type	Value Range/Format	Description
name	String	1 to 32 character long string.	The name of internal program as configured in RPS.

Name	Type	Value Range/Format	Description
siid	String	String in the form of **A.B.<String>.C.D** Where A, B, C and D are integer values	Auto generated unique Identifier.
deviceList	Array of String	Array of strings in the form of **A.B.<String>.C.D** Where A, B, C and D are integer values	List of SIIDs of all devices that are configured for the internal program

Device configuration object structure:

Name	Type	Value Range/Format	Description
name	String	1 to 32 character long string.	The name of device as configured in RPS.
siid	String	String in the form of **A.B.<String>.C.D** Where A, B, C and D are integer values	Auto generated unique Identifier.
type	String	Device type in format <type>.<subtype>	**Type:** POINT **Subtypes:** ONBOARD,TAMPER, LSNEXPANDER, PIR, SEISMIC, GLASSBREAK, CONTACT, PANIC, UNIVERSAL, MCP, VIRTUAL **Type:** FIREDETECTOR **Subtype:** O, OT, OC, OTC, T **Type:** OUTPUT **Subtype:** CONTROL, LED, SIREN,STROBE, KPSPEAKER (Keypad Speaker), VIRTUAL **Type:** LSNBUS **Subtype:** LOOP,STUB **Type:** KEYSWITCH

Name	Type	Value Range/Format	Description
			Subtype: STATIC,DYNAMIC, VIRTUALSTATIC, VIRTUALDYNAMIC **Type:** COMMUNICATOR **Subtype:** AT2000, AT3000 **Type:** COUPLER **Subtype:** - **Type:** LSNAUX **Subtype:** - **Type:** GROUNDFAULT **Subtype:** - **Type:** BATTERYCHARGER **Subtype:** - **Type:** BATTERY **Subtype:** - **Type:** MAINS **Subtype:** - **Type:** PSCANOUTPUT **Subtype:** - **Type:** POWERDEVICE **Subtype:** - **Type:** PRINTER **Subtype:** - **Type:** BLOCKLOCK **Subtype:** - **Type:** SMARTKEY **Subtype:** -

Name	Type	Value Range/Format	Description
			Type: DEMODULE **Subtype:** - **Type:** POWERSUPPLY **Subtype:** - **Type:** LSNGATEWAY **Subtype:** - **Type:** KEYPAD **Subtype:** - **Type:** OII **Subtype:** - **Type:** SUPERV **Subtype:** BIS,IPC,OII
partOfWalktest	Bool	true/false	Indicates whether this device is part of a walktest. If true, the resource will implement the walktest interface.
bypassable	Bool	true/false	Indicates whether device can be bypassed. If true, the resource implements the bypass interface.
parentSiid	String	String in the form of A.B.<String>.C.D Where A, B, C and D are integer values.	Parent Device SIID. This will be "<Empty>" if no parent Device. This would indicate where the device is configured. For eg. On board, internal CAN etc

5.1 *Status Retrieval*

Request

Verb	*GET*
Query Parameters	(none)

Response (Success)

Return Code	200
Content	Configuration Object Structure

Response (Error)

See standard errors defined in OII Base Specification.

6 Panel

The panel resource provides information on the current status of the panel, for e.g. panel being in failsafe, in installer mode, last configuration update date time and last database update date and time. The resource will allow an authorized client to restart the panel.

Overview

OII Type	Supported Operations
IN.panel.1.1	Status Retrieval Restart

Object Structure

Name	Type	Value Range/Format	Description
@type	Array of Strings	[IN.panel.1.1]	Fixed type identifier
@self	String	URL starting with "/"	Link to the current resource
failsafe	Bool	True/False	Indicates if the panel is in failsafe
installerMode	Bool	True/False	True if Panel is in installer mode, else false
cfgStatus	String	"default" "latest" "last_good"	The configuration status of the panel – "default" – Panel is running default configuration "latest" – Panel is running the latest configuration sent to it

Name	Type	Value Range/Format	Description
			“last-good” – Panel is running the last good configuration
lastCfgUpdt	String	Time and date in OII Base specification compliant format (e.g. 2014-07-20T14:11:42Z, 2014-07-20T16:11:42+02:00)	Date and time of the last configuration update on the panel
lastUserDbUpdt	String	Time and date in OII base specification compliant format (e.g. 2014-07-20T14:11:42Z, 2014-07-20T16:11:42+02:00)	Date and time of the last user database update on the panel from RPS (remote configuration tool)
isPanelLoaded	Bool	True/False	Indicates the panel is in a temporary busy state (resulting in possible delays for responses over OII. No data loss is expected.)
restartImminent	Bool	True/False	Indicates that the panel is overloaded, due to which the panel will restart.

6.1 *Status Retrieval*

Request

Verb	*GET*
Query Parameters	(none)

Response (Success)

Return Code	200
Content	Panel Object Structure

Response (Error)

See standard errors defined in OII Base Specification.

6.2 *Restart*

It is possible to restart the panel over the interface. The client can specify if information persisted on the panel needs to be retained across the restart. Only an authorized client can issue a restart command on the panel.

Request

Verb	*POST*
Query Parameters	(none)

Request Body:

Name	Type	Value Range/Format	Description
@cmd	String	RESTART	Fixed String identifier for this command
persistData	BOOL	True/False	If false, Panel will erase all data that is persisted. If true, Panel will retain all data that is persisted

Response (Success)

Return Code	202
Content	-

Response (Error)

See standard errors defined in OII Base Specification.

7 Time

The time resource provides information on the current system time of the map including time zone. A client can also use this resource to set the time of the panel. The time information is in compliance with the time stamp definition in the OII base specification. The time is given in precision of milliseconds.

Overview

OII Type	Supported Operations
IN.time.1	Status Retrieval Set time

Object Structure

Name	Type	Value Range/Format	Description
@type	Array of strings	[IN.time.1]	Fixed type identifier
@self	String	URL starting with "/"	Link to the current resource
timeZone	String	Time zone string	Time zone following the IANA Time Zone Database (also known as Olson database) name e.g. Asia/Kolkata, Europe/Berlin
dateTime	String		Time and date in OII base specification compliant format including milliseconds (e.g. 2014-07-20T14:11.345:42Z, 2014-07-20T16:11:42.345+02:00). A client can extract local as well as UTC date time from this string.

7.1 *Status Retrieval*

Request

Verb	*GET*
Query Parameters	(none)

Response (Success)

Return Code	200
Content	Time object structure

Response (Error)

See standard errors defined in OII Base Specification.

7.2 *Set time*

It is possible to set the date and time of the panel over the OII. This can be used for commissioning purposes or for time synchronization, where a client sets the time in regular intervals to assure a desired synchronization of its internal clock to the clock of the MAP.

Currently the time zone is only configurable via RPS and cannot be set via the OII. This is considered to be appropriate, as the MAP needs to be configured over RPS and the time zone should not change over time, as the installation location of the MAP is fixed.

The time that is provided in the request is interpreted as UTC time. This means, even if the date time is including time zone information e.g. 2014-07-20T16:11:42+02:00 the actual UTC time is calculated from that e.g. 2014-07-20T14:11:42Z and the internal clock's UTC is set accordingly. When fetching the time using a GET, the local time reflecting the configured time zone will be provided as described in Sec. 7.1.

Request

Verb	*POST*
Query Parameters	(none)

Request Body:

Name	Type	Value Range/Format	Description
@cmd	String	SETTIME	Fixed String identifier for this command
utcDateTime	String	Date as defined in OII base specification including timezone but in second precision (e.g. 2014-07-20T16:11:42+02:00). Date between 1970-01-01 and 2036-12-31.	If no time zone is specified (e.g. +02:00 or Z), the request will be rejected.

Response (Success)

Return Code	202
Content	-

Response (Error)

See standard errors defined in OII Base Specification.

8 User

This resource models a MAP user. It provides information about its id, activation status and allows activation/deactivation of the user. In addition, a user list exists, which represents the list of all users configured in the MAP.

Overview

OII Type	Supported Operations
IN.user.1	Status Retrieval Activate ActivateUntil Deactivate

Object Structure

Name	Type	Value Range/Format	Description
@type	Array of strings	["IN.user.1"]	Fixed type identifier
@self	String	URL starting with "/"	Link to the current resource
id	String	3 digit user ID ranging from 004 to 999.	The unique ID of the user
active	Boolean	true/false	True, if the user is currently active. False, if user is deactivated. Only active users can interact with the system via e.g. OII, system keypad, smart key
activeUntil	String	Date as defined in OII base specification (i.e. YYYY-MM-DD). Date between 1970-01-01 and 2036-12-31.	Date until when the user will be active. Empty string if user is indefinitely active
activeFrom	String	Date as defined in OII base specification Date between 1970-01-01 and 2036-12-31.	Date from when the user is active

Name	Type	Value Range/Format	Description
type	String	“endUser:Standard”, “endUser:Temporary”, “endUser:OneTimeUser”, “ installer:Standard”, “ installer:OneTimeUser”, “ installer:Temporary”	A string representing the type and subtype of users Installer: allowed to access the panel only when panel is in installer mode endUser : allowed to access the panel only when panel is not in installer mode (normal operation) Standard: User is indefinitely active until deactivated explicitly One Time User : User who is allowed to login only once on the keypad Temporary: User is active for a limited period - between the dates specified in “activeFrom” to “activeUntil”
name	String	E.g. John Doe	User name as FirstName LastName
language	String	“en-US”, “de-DE”, “fr-FR”, “nl-NL”, “hu-HU”, “pl-PL”, “it-IT”, “ru-RU”, “es-ES”, “cs-CZ”, “pt-PT”,	Language of the user. All menus would appear in the language selected when user logs into the keypad en-US – American English de-DE - German fr-FR - French nl-NL - Dutch hu-HU - Hungarian

Name	Type	Value Range/Format	Description
		"lv-LV" "ro-RO" "lt-LT"	pl-PL - Polish it-IT - Italian ru-RU - Russian es-ES - Spanish cs-CZ - Czech pt-PT – Portuguese lv-LV – Latvian ro-RO – Romanian lt-LT - Lithanian
passcodeChngRqd	Boolean	true/false	True if the user is forced to change his passcode on first login on the keypad else false
accessProfileName	String		Name of the access profile configured for the user
smartkeyProfileName	String		Name of the smartkey profile configured for the user
isOiiUser	Boolean	true/false	true if the user is allowed to access the panel over the open intrusion interface
isOiiUserKpUser	Boolean	true/false	True if the user configured as an OII User is also allowed to login on the keypad

8.1 ***Status Retrieval***

Request

Verb	*GET*
Query Parameters	(none)

Response (Success)

Return Code	200
Content	User object structure

Response (Error)

See standard errors defined in OII Base Specification.

8.2 *Activate*

This command initiates activation of a user. The command can be used to activate users of type endUser:Standard , endUser:OneTimeUser. The user will be activated with immediate effect.

In case activate is called on a endUser:Temporary a 403 Forbidden will be returned. For activation of temporary users the command "activate until" must be used. In case activate is called on any installer type user a 403 Forbidden will be returned. Installer users cannot be activated or deactivated.

Request

Verb	*POST*
Query Parameters	(none)

Request Body:

Name	Type	Value Range/Format	Description
@cmd	String	ACTIVATE	Fixed String identifying this command

Response (Success)

Return Code	204
Content	-

Response (Error)

See standard errors defined in OII Base Specification.

8.3 *Activate Until*

This command initiates activation of a temporary user. The command can be used to activate users of type endUser:Temporary. The user will be activated with immediate effect. When activating a temporary user a date needs to be specified until when the user shall be kept active.

In case activate until is called on a endUser:Standard or endUser:OneTimeUser a 403 "Forbidden" will be returned. For activation of endUser:Standard and endUser:OneTimeUser the command “activate” must be used. In case activate until is called on any installer type user a 403 Forbidden will be returned. Installer users cannot be activated or deactivated.

Request

Verb	*POST*
Query Parameters	(none)

Request Body:

Name	Type	Value Range/Format	Description
@cmd	String	ACTIVATEUNTIL	Fixed String identifying this command
activeUntil	String	Date in YYYY-MM-DD format between 1970-01-01 and 2036-12-31	Date until which the user will be active

Response (Success)

Return Code	204
Content	-

Response (Error)

See standard errors defined in OII Base Specification.

8.4 *Deactivate*

This command initiates the deactivation of a user of type endUser:Standard, endUser:Temporary and endUser:OneTimeUser. The user will be deactivated with immediate effect.

Deactivation of installer users is not supported. A 403 Forbidden will be returned in case command is requested on any type of installer user.

Request

Verb	*POST*
Query Parameters	(none)

Request Body:

Name	Type	Value Range/Format	Description
@cmd	String	DEACTIVATE	Fixed String identifying this command

Response (Success)

Return Code	204
Content	-

Response (Error)

Return Code	Content	Description
409 (Conflict)	User logged in on a System keypad	Response if the request is to deactivate a user who is locked since the user is logged in on a system keypad
409 (Conflict)	Cannot deactivate yourself	Response if a user attempts to deactivate himself/herself

9 User List Resource

The user list resource allows batched access to multiple or all users of the panel. A client can also use this resource to activate or deactivate multiple users with a single request.

Overview

OII Type	Supported Operations
IN.userList.1, OII.list.1	Status Retrieval Activate ActivateUntil Deactivate

Object Structure

Name	Type	Value Range/Format	Description
@type	Array of strings	[IN.userList.1, OII.List.1]	Fixed type identifier
@self	String	URL starting with "/"	Link to the current resource

Name	Type	Value Range/Format	Description
list	Array	Array of User Objects of type IN.user.1	

9.1 *Status Retrieval*

Get the user resources selected by the specified filters.

Request

Verb	*GET*		
Query Parameters	url	Resource ID as specified for OII.list.1	optional

Response (Success)

Return Code	200
Content	UserList object structure. The list object will only contain resources which match the defined filters given in the query parameters.

Response (Error)

See standard errors defined in OII Base Specification.

9.2 *Activate*

This command activates a set of users selected by the specified filters.

Activate command can be used only on users of type endUser:Standard and endUser: OneTimeUser. For activation of type endUser:Temporary, Activate Until command will have to be used. Installer type users cannot be activated. The response code will depend on whether the atomic flag is specified.

Request

Verb	*POST*		
Query Parameters	url	Resource ID as specified for OII.list.1	optional
	atomic	If set, command is only executed when execution on selected resources is possible.	optional

Request Body:

Same as for IN.user.1

Response (Success)

Return Code	204
Content	-

Response (Error)

Return Code	Content	Description
409 (Conflict)	Comma separated list of urls	List of resource urls which prevented the command from being executed when atomic is true
409 (Conflict)	Not executed	String describing that application rules prevented execution of the command on any element of the list. Will be only returned when atomic is false.

9.3 *Activate Until*

This command activates a set of users selected by the specified filters.

Activate Until command can be used only on users of type endUser:Temporary. For activation of type endUser:Standard and endUser:OneTimeUser, Activate command will have to be used. Installer type users cannot be activated. The response code will depend on whether the atomic flag is specified.

Request

Verb	*POST*		
Query Parameters	url	Resource ID as specified for OII.list.1	optional
	atomic	If set, command is only executed when execution on selected resources is possible.	optional

Request Body:

Same as for IN.user.1

Response (Success)

Return Code	204
Content	-

Response (Error)

Return Code	Content	Description
409 (Conflict)	Comma separated list of urls	List of resource urls which prevented the command from being executed when atomic is true
409 (Conflict)	Not executed	String describing that application rules prevented execution of the command on any element of the list. Will be only returned when atomic is false.

9.4 *Deactivate*

This command deactivates a set of users selected by the specified filters. Installer users cannot be deactivated.

Note: A user cannot deactivate himself/herself. Hence if the filter specified includes the user who is requesting the command, that user will be ignored.

Request

Verb	*POST*		
Query Parameters	url	Resource ID as specified for OII.list.1	optional
	atomic	If set, command is only executed when execution on selected resources is possible.	optional

Request Body:

Same as for IN.user.1

Response (Success)

Return Code	204
Content	-

Response (Error)

Return Code	Content	Description
409 (Conflict)	Comma separated list of urls	List of resource urls which prevented the command from being executed when atomic is true
409 (Conflict)	Not executed	String describing that application rules prevented execution of the command on any element of the list. Will be only returned when atomic is false.

10 Incident

Alarms and troubles of the MAP5000 are modelled as so called incidents. In difference to other resources like area or point, an incident resource will not always exist, but will be created when an alarm/trouble appears in the system. The incident resource will also be removed from the OII when the alarm/trouble is resolved and disappears from the keypad.

The incident resource itself contains all relevant information about the alarm. In particular, it conveys who has acknowledged or handled the alarm. Creation, deletion and state changes of an Incident are communicated via events.

The incidents are located under the root resource “/inc”. This resource acts as an incident list, where all pending incidents in the system can be accessed. The individual incidents are located under /inc with the following structure:

/inc/[Area ID]/[incident ID]

The fields are as follows:

- **Area ID:** Complete area SIID (e.g. 1.1.Area.2.3)
- **Incident ID:** Unique ID of an incident in the given area. A client shall treat this ID as an opaque string and shall not expect particular semantics or sequence information from the ID. In case the panel is rebooted while incidents are detected, the panel will detect the incidents as new incidents in the system. The ids will be chosen uniquely so that there is no conflict in the ids over a reboot.

Depending on the device type and incident type, the life cycle of the incident varies. The interface provides information on what the client is expected to do. The “Handling Required” attribute of the incident resource indicates whether a user needs to handle/acknowledge the incident via the OII or the keypad before it can get resolved and deleted (See Fig 2).

Usually incident can only be handled when area is disarmed. However there is a configuration option to explicitly allow handling in armed areas ("Allow Reset of Armed Areas").

Some types of incidents (e.g. LSN Loop tamper) require to be handled after the device is back to normal thus "Handling Required" will remain true even if it was already handled. User has to retry handling until the incident is deleted (see Fig 3).

Other types of incidents (e.g. AC Mains fault) do not require handling and will be deleted once the device reports a normal state. A user can still handle such incidents but it does not have any influence on the lifecycle of the incident (see Fig 4)

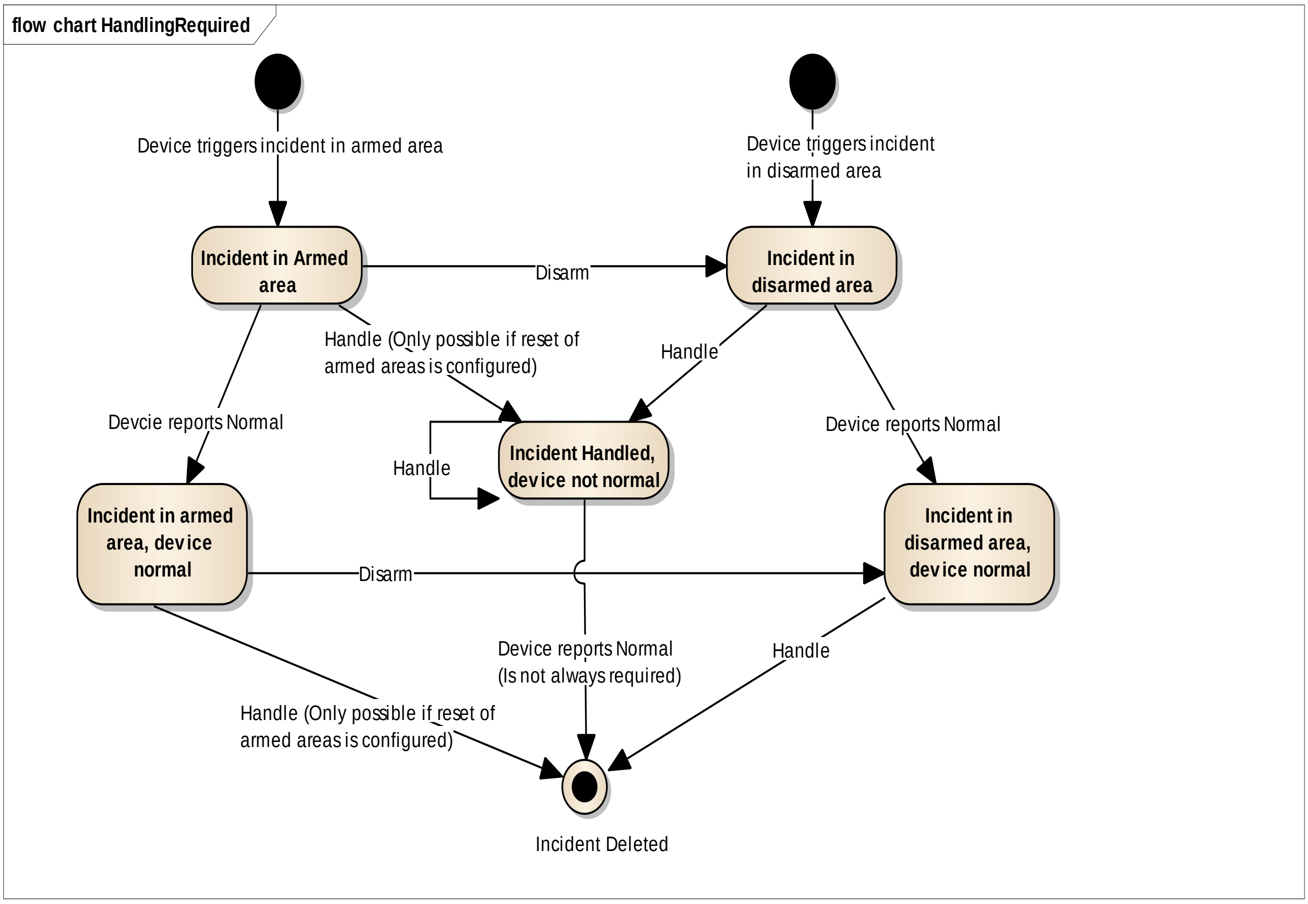

Fig. 2 Incident requiring handling

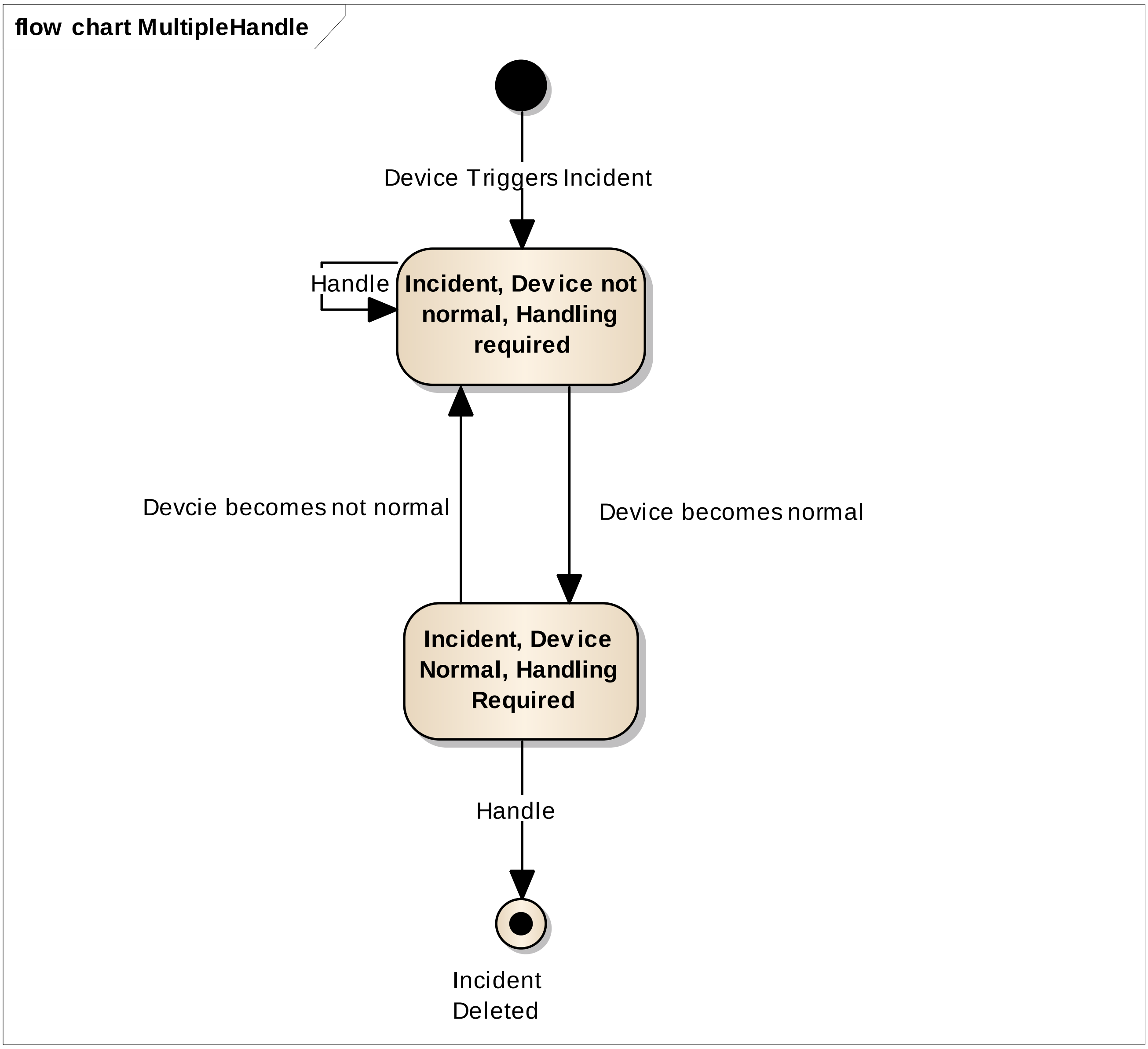

Fig. 3 Incident requiring multiple handling

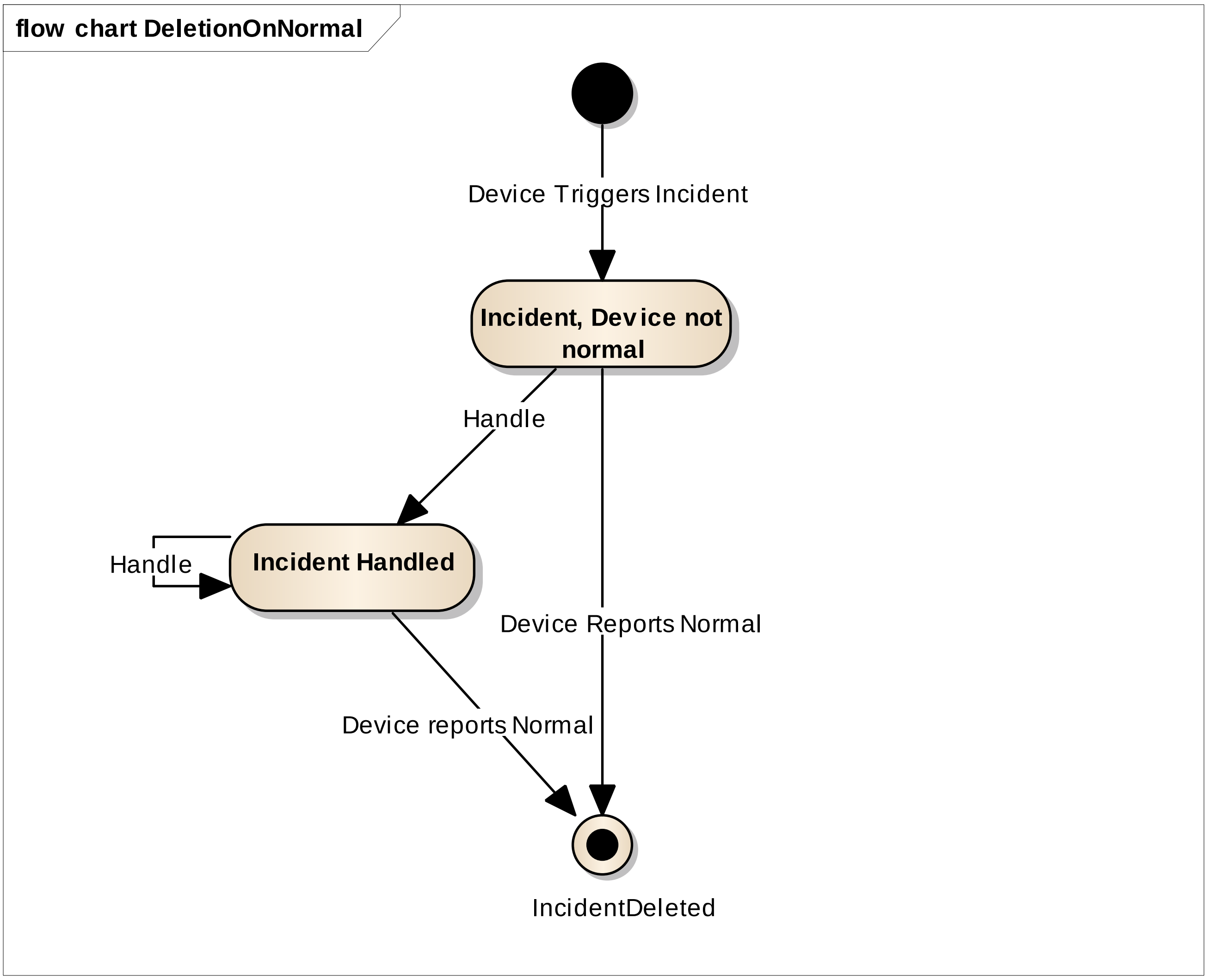

Fig. 4 Incident requiring no handling

Incidents are categorized as follows:

[type].[category].[sub-category]

- **type**: Gives information about the severity of the incident. Possible values are:
  - Trouble: indicates malfunction of the system
  - Alarm: indicates an alarm in the system
- **category:** Defines the function set of the panel that is effected by the incident:
  - Intrusion: The incident relates to intrusion detection functionality of the panel.
  - Fire: The incident relates to fire detection functionality of the panel.
  - System: The incident relates to the function of the system (particularly system components like panel, power supply, battery, etc.)
  - Technical: The incident relates to components in the system which are not intrusion, fire or system components.

- **sub-category:** A precise description of the state detected in the system. (e.g. Battery Low, Printer Cover Open, Hold-up)

10.1 *Incident Resource*

An incident resource represents a single alarm or trouble. The incident resource is generic, in the sense that any type of incident, alarm or trouble, will share the data and operations of the incident type. For each particular alarm or trouble a dedicated, specialized type will be defined, that will allow a client to understand what type of alarm or trouble it is (e.g. intrusion alarm, duress, etc.). A specialized incident may add additional information items or operations.

Overview

OII Type	Supported Operations
IN.inc.1.1	Status Retrieval Handle Silence

Object Structure

Name	Type	Value Range/Format	Description
@type	Array of strings	[IN.inc.1.1]	Fixed string identifier
@self	String	URL starting with "/"	Link to this resource
incType	String	String of the format - [type].[category].[sub-category]	e.g. Alarm.Intrusion.General, Trouble.System.PrinterPaperLow
extInc	Bool	True/False	Indicates whether the incident is an external incident or an internal incident (external incidents are in general incidents that are reported to the central station)
time	String	OII timestamp	Time when the incident occurred in OII compliant format in precision of seconds.

Name	Type	Value Range/Format	Description
handlingState	String String String	Triple of “state”, “user”, “interface”	• State: NONE/ACKED/HANDLED (MAP currently supports only NONE and HANDLED) • User: the MAP user that handled the incident • Interface: the interface that was used to handle the incident (i.e. OII or KEYPAD)
handlingRequired	Bool	true/false	Indicates whether the incident requires handling by the user in order to be resolved. If false, the incident will get deleted once the reason for the incident is not present anymore.
triggeredBy	String	url	The id of the element that triggered the incidents (e.g. ID of the door contact that triggered an intrusion alarm)
relatesTo	Array	Array of urls	Reference to other elements that are affected by the alarm. Will at least contain the url of the area in which the incident occurred.
silenced	Bool	true/false	Indicates whether the outputs and sirens configured to sound for this incident are turned on or off.
counter	Number		Number of times the incident occurred. Value will usually be one, as an incident is normally triggered only once. In case of a duress alarm, the counter will increase with every activation of a duress alarm from the same location (pressing the duress button multiple times)

Example

```
{
  "@type": [ "IN.inc.1" ],
  "@self": "/inc/1.1.Area.2.4/234",
  "time": "2014-07-20T16:11:42+02:00",
  “handlingState":
   {
      “state”: “NONE”,
      “user”:””,
      “interface”:””
   }
  “triggeredBy”: "/1.1.Point.5.23",
  "relatesTo": [
    "/1.1.Area.2.4"
  ],
  “counter”: 1
}
```


10.1.1 Status Retrieval

Request

Verb	*GET*
Query Parameters	(none)

Response (Success)

Return Code	200
Content	Incident object structure

Response (Error)

See standard errors defined in OII Base Specification.

10.1.2 Handle

Request

Verb	*POST*
Query Parameters	(none)

Request Body:

Name	Type	Value Range/Format	Description
@cmd	String	HANDLE	Handling will result in the incident disappearing in case the reason for the incident disappeared.

Response (Success)

Return Code	202
Content	-

Response (Error)

Return Code	Content	Description
409 (Conflict)	Area armed	Response if the incident being handled was detected in an area which is armed. Area needs to be disarmed before the incident can be handled.

In addition see standard errors defined in OII Base Specification.

10.1.3 Silence

Request

Verb	*POST*
Query Parameters	(none)

Request Body:

Name	Type	Value Range/Format	Description
@cmd	String	SILENCE	Deactivates the sirens or outputs configured to sound when this incident occurs. It is not possible to reactivate the sirens and outputs through a command on the incident.

Response (Success)

Return Code	202
Content	-

Response (Error)

See standard errors defined in OII Base Specification.

10.2 *Incident Categorization*

The following table provides an overview of the types of troubles the MAP5000 is supporting. In case a particular type adds additional information items or operation, a detailed specification will be given in a dedicated section.

Alarms indicate significant situation has been detected in the system that requires immediate action (e.g. intrusion alarm, fire alarm).
Troubles indicate that the system is not fully operational and action may be advised (e.g. device in an unarmed area malfunctioning).

10.2.1 Troubles

10.2.1.1 Intrusion

Type	Category	Sub-Category	Description
Trouble	Intrusion	General	Malfunction of an intrusion device
Trouble	Intrusion	Antimask	When a PIR detector is covered or masked.

10.2.1.2 Fire

Type	Category	Sub-Category	Description
Trouble	Fire	General	Malfunction of a fire detector
Trouble	Fire	Optical	Malfunction of the optical sensor of the fire detector
Trouble	Fire	Chemical	Malfunction of the chemical detector of the fire sensor
Trouble	Fire	Thermal	Malfunction of the thermal sensors of the fire detector
Trouble	Fire	Output Supervisory	Malfunction of a supervised output device

10.2.1.3 Technical

Type	Category	Sub-Category	Description
Trouble	Technical	General	Malfunction of a technical detector

10.2.1.4 System

Type	Category	Sub-Category	Description
Trouble	System	General	Malfunction of a general system device (Specific device would detect specific malfunction incidents)
Trouble	System	Initialization	Incident detected when an element in the system went into initialization state when it was not expected.
Trouble	System	Failure	Incident detected when there is a system failure
Trouble	System	Communication	Malfunction of a communication device like AT2000 communicator
Trouble	System	Battery Low	The battery connected to the power supply is low on power
Trouble	System	Battery failure	The battery connected to the power supply has failed
Trouble	System	Battery Missing	The battery configured in the system is not connected
Trouble	System	Battery Self Test Failed	Self Test of the battery failed
Trouble	System	Main Power Failure	Failure in the main power to the power supply

Type	Category	Sub-Category	Description
Trouble	System	Ground fault	Fault detected from a Power supply due to Improper Grounding
Trouble	System	Over Current	Incident indicating that more power is being drawn from the device than the expected limits
Trouble	System	Output Fault	Malfunction of a supervised output device
Trouble	System	Printer Paper Low	Printer is low on paper
Trouble	System	Printer Cover Over	Printer cover is open
Trouble	System	Device Firmware Corrupt	Firmware on the peripheral device is corrupted Detected for system keypad, gateway, DE Module and the power supply
Trouble	System	Device Firmware Programming	Firmware on the peripheral device is being updated. Detected for system keypad, gateway, DE Module and the power supply
Trouble	System	Management System Interface	Management System monitoring the panel has lost connection to the panel

10.2.2 Alarms

10.2.2.1 Intrusion

Type	Category	Sub-Category	Description
Alarm	Intrusion	General	Alarm from any Intrusion Detector
Alarm	Intrusion	Hold-up	Alarm from a detector under panic situation
Alarm	Intrusion	Amok	Alarm from a detector under amok situation
Alarm	Intrusion	Duress	Alarm from a detector under threat
Alarm	Intrusion	Door Not Locked	Alarm when a door is not locked properly, even after Exit Delay Expiry
Alarm	Intrusion	Exit Error	Alarm in case of un-authorised presence in premises, even after Exit Delay expiry.

10.2.2.2 Fire

Type	Category	Sub-Category	Description
Alarm	Fire	General	Alarm from any Fire detector

10.2.2.3 Technical

Type	Category	Sub-Category	Description
Alarm	Technical	General	Alarm from any Technical detector

10.2.2.4 System

Type	Category	Sub-Category	Description
Alarm	System	General	Alarm from any system device
Alarm	System	User Code Tamper	Invalid PIN entry consecutively multiple times
Alarm	System	Tamper	Sabotage Alarm
Alarm	System	Output	Alarm from a supervised output

11 Incident List

The IncidentList resource wraps the list of all incidents in the panel, to provide status of and access to multiple incidents at the same time.

Overview

OII Type	Supported Operations
IN.incList.1, OII.list.1	Status Retrieval Handle Silence

Object Structure

Name	Type	Value Range/Format	Description
@type	Array of strings	[IN.incList.1, OII.list.1]	Fixed type identifier
@self	String	URL starting with "/"	Link to the current resource
list	Array	Array of Area Objects of type IN.inc.1.1	

Example


```
{
  "@type": ["IN.incList.1","OII.list.1"],
  "@self": "/inc",
  "list": [
    {
  "@self": "/inc/1.1.Area.2.4/234",
  "@type": [ "IN.inc.1" ],
  "time": "2014-07-20T16:11:42+02:00",
  "handlingState":
    {
      "state": NONE
      "user":"",
      "interface":""
    }
   "triggeredBy": "/1.1.Point.5.23",
  "relatesTo": [  "/1.1.Area.2.4"  ],
  "counter": 1
  }
  ]
}
```


11.1 *Status Retrieval*

The incident list allows retrieval of status information of multiple incidents. The incidents that are to be fetched are selected using query parameters (url). If no parameters are specified, the status of all incidents is provided.

Request

Verb	*GET*		
Query Parameters	url	Resource ID as specified for OII.list.1	optional

Response (Success)

Return Code	200
Content	IncList object structure. The list object will only contain resources which match the defined filters given in the query parameters.

Response (Error)

See standard errors defined in OII Base Specification.

11.2 *Handle*

The Incident List allows handling of multiple incidents with a single request. The incidents that are to be handled are selected using query parameters (url). Thus, the client can use the IncList resource to handle one, multiple or all incidents in the panel. As the incidents are grouped also by the areas, all incidents of a given area can be handled by specifying a filter with url=/inc/[AREA SIID]/* (e.g. url=/inc/1.1.Area.2.4/*).

Request:

Verb	POST		
Query Parameters	url	Resource ID as specified for OII.list.1	optional
	atomic	If set, command is only executed when execution on selected resources is possible.	optional

Request Body:

Same as for IN.inc.1.1

Response (Success)

Return Code	202
Content	-

Response (Error)

Return Code	Content	Description
409 (Conflict)	Comma separated list of urls	List of resource urls which prevented the command from being executed when atomic is true
409 (Conflict)	Not executed	String describing that application rules prevented execution of the command on any element of the list. Will be only returned when atomic is false.

Additionally see standard errors defined in OII Base Specification.

11.3 *Silence*

The Incident List allows silencing of multiple incidents with a single request. The incidents that are to be handled are selected using query parameters (url). Thus, the client can use the IncList resource to handle one, multiple or all incidents in the panel. As the incidents are grouped also by areas, all incidents of a given area can be handled by specifying a filter with url=/inc/[AREA SIID]/* (e.g. url=/inc/1.1.Area.2.4/*).

Request

Verb	POST		
Query Parameters	url	Resource ID as specified for OII.list.1	optional
	atomic	If set, command is only executed when execution on selected resources is possible.	optional

Request Body:

Same as for IN.inc.1.1

Response (Success)

Return Code	202
Content	-

Response (Error)

Return Code	Content	Description
409 (Conflict)	Comma separated list of urls	List of resource urls which prevented the command from being executed when atomic is true
409 (Conflict)	Not executed	String describing that application rules prevented execution of the command on any element of the list. Will be only returned when atomic is false.

12 Network Settings

This resource allows inspecting and modifying the IP network settings of the MAP.
User attempting to modify the network settings of the MAP should have the appropriate permissions.

Overview

OII Type	Supported Operations
IN.network.1	Status Retrieval Activate DHCP Set Static IP Address

Object Structure

Name	Type	Value Range/Format	Description
@type	Array of strings	[IN.network.1]	Fixed type identifier
@self	String	URL starting with "/"	Link to the current resource
dhcp	Bool	true/false	True means that the panel is fetching its IP address via DHCP. False means the IP address is statically configured.
ipv4Address	String	IPv4 Address	Current IP address of panel
ipv4Netmask	String	IPv4 Subnet mask	Current subnet mask used by the panel

Name	Type	Value Range/Format	Description
ipv4Gateway	String	IPv4 Address	Current default gateway used by the panel

Example


```
{
  "@type": ["IN.network.1"],
  "@self": "/network",
  "dhcp": true,
  "ipv4Address ": "192.168.1.11",
  "ipv4Netwask": "255.255.255.0",
  "ipv4Gateway": "192.168.1.1"
}
```


12.1 *Status Retrieval*

Request

Verb	*GET*
Query Parameters	(none)

Response (Success)

Return Code	200
Content	Network object structure

Response (Error)

See standard errors defined in OII Base Specification.

12.1 *Activate DHCP*

This operation activates the use of DHCP. In order to deactivate DHCP a static IP address needs to be configured using “Set Static IP Address”. After sending the response, the panel reboots to apply the new settings.

Request

Verb	POST
Query Parameters	-

Request Body

Name	Type	Value Range/Format	Description
@cmd	String	ACTIVATEDHCP	

Response (Success)

Return Code	202
Content	-

Response (Error)

See standard errors defined in OII Base Specification.

12.2 *Set Static IP Address*

This operation sets the IP Address of the panel, deactivates DHCP (if previously activated). After sending the response, the panel reboots to apply the new settings.

Request

Verb	POST
Query Parameters	-

Request Body:

Name	Type	Value Range/Format	Description
@cmd	String	SETIPV4	
ipv4Address	String	IPv4 Address	Current IP address of panel
ipv4Netmask	String	IPv4 Subnet mask	Current subnet mask used by the panel
ipv4Gateway	String	IPv4 Address	Current default gateway used by the panel

Response (Success)

Return Code	202
Content	-

Response (Error)

See standard errors defined in OII Base Specification.

13 History

The history resource provides access to the MAP history data base. Each entry in the history describes an event in the panel that has been configured to be logged in the history. As the history consists of a large amount of entries, dedicated filters are defined that allow paged access to the history. Thus, a client does not need to fetch the complete history in one GET request. Due to its size, the history data resource does not create any events.

MAP provides 2 separate history data – main history and IPC history. Main history consists of events in the panel configured to be logged. IPCHistory (IP Communicator History) consists only of events related to reporting over IP. (IPCHistory is applicable only if the panel has been configured to report over IP. Events in IPCHistory will not be logged in panel main history). A Client will have to fetch events from main history and IPCHistory separately.

The history in the MAP is implemented as a ring buffer causing older events to be deleted when the buffer is full. Thus, a client that wants to archive all events need to read out the history regularly to make sure that no events get overwritten before they could be fetched.

Each history entry is following a fixed structure of:

- Event id: Unique ID for this event
- Timestamp: Time when the event occurred. Timestamp is ISO8601 compliant but not OII date time compliant as time zone is not included.
- Event name: Human readable string describing the event
- SIID: Id of the system element that triggered that event
- Parameters: Additional information on the event in a key value fashion. Each key and value are separated by a " = " (two whitespaces + =). Multiple key value pairs are separated by a "\n", which is a JSON reserved character. For example, the keys IsMandatory and ModifierUser_ID are represented as

IsMandatory = No\nModifierUser_ID = 2\n

The history resource is represented as a JSON structure which contains an array of history entries. Each history entry contains the above defined fields separated by a semicolon ";". For example:

"883953013;2014-07-24T13:24:13.256;RPS Accessed the System;TA_OCCUR;1.1.System.4.1;IsMandatory = No\n"

Overview

OII Type	Supported Operations
IN.history.1	Status Retrieval

Object Structure

Name	Type	Value Range/Format	Description
@type	Array of strings	["IN.history.1"]	Fixed type identifier
@self	String	URL starting with "/"	Link to the current resource
history	Array of Strings		Contains one string for each event in the event history. The content of this field depends on the selected filters for the query

13.1 *Status Retrieval*

Request

Verb	*GET*		
Query Parameters	fromeid	Starting history event id to be retrieved from	optional
	maxevents	Max history events to be sent in response	optional

Response (Success)

Return Code	200
Content	History object structure with the history element containing only entries which match the specified filter

Response (Error)

If the panel is processing a history request already when there is another request for History retrieval, the panel shall return a 503 (Service Unavailable) response code. A client can try again at a later point of time.

14 IPC History

IPC history has the same interface as history (i.e. IN.history.1). In order to distinguish IPC and regular history, the history type IN.ipcHistory.1 is introduced. This type does not add any particular functionality.

Overview

OII Type	Supported Operations
IN.history.1 IN.ipcHistory.1	Status Retrieval

Object Structure

same as IN.history.1

14.1 *Status Retrieval*

same as IN.history.1

15 Area

The area resource represents an individual area that is configured with the MAP. Clients that are not aware of the available areas can use the AreaList resource (see below), to inspect the areas that are configured with MAP. In the current version of the specification each area is accessible at a location corresponding to its SIID. For example, an area with the SIID 1.1.Area.2.5 is accessible under /1.1.Area.2.5.

The area resource itself provides information about the current area state including whether it can be armed or disarmed. Further, arming and disarming of an area is supported, as well as control over test modes and the chime mode.

Overview

OII Type	Supported Operations
IN.area.1	Status Retrieval Arm Disarm Start Walktest Stop Walktest Start Motion Detector Test Mode Stop Motion Detector Test Mode

OII Type	Supported Operations
	Start Chime Mode Stop Chime Mode Start Bell Test Arming Info

Object Structure

Name	Type	Value Range/Format	Description
@type	Array of strings	[“IN.area.1”]	Fixed type identifier
@self	String	URL starting with “/”	Link to the current resource
armed	Bool	true/false	Indicates whether panel is armed
transitionalState	String	INENTRYDELAY INEXITDELAY <EmptyString>	An empty JSON string (i.e. “”) indicates that area is not in a transitional state at the moment
oiiArmable	Bool	true/false	True, if it is possible to dis-/arm the area over the interface. Areas may be configured only to be dis-/armed blocklocks, thus no dis/-arm will be allowed over the interface.
readyToArm	Bool	true/false	If true, arming of the area is possible. Arming can either be done by a user using MAP peripherals e.g. Keypad, smartkey or via the OII, if area is “OII Armable”.

Name	Type	Value Range/Format	Description
readyToDisarm	Bool	true/false	If true, disarming of the area is possible. Disarming can either be done by a user using MAP peripherals e.g. Keypad, smartkey or via the OII, if area is “OII Armable”.
numberOfBypassedDevices	Number	0 – 1500	Number of devices that are bypassed in that area
walktest	String	Empty String or link to running walktest	If empty, than no walktest is running for this area. Otherwise this field contains the link to the active walktest relative to the server base URL.
motionDetectorTestActive	Bool	true/false	
chimeModeActive	Bool	true/false	
incs	Array	List of links to currently pending incidents that relate to this area.	This field shows the relationship between incidents (alarm/trouble) and an individual area. Details about the incident are contained in the incident resource at its URL.

Example:

```
{
  "@type": ["IN.area.1"],
  "@self": "/1.1.Area.1.1",
  "armed":false,
  "transitionalState":"",
  "readyToArm":true,
  "readyToDisarm":false,
  "oiiArmable":true,
  "numberOfBypassedDevices":2,
  "walkTestActive": false,
  "motionDetectorTestActive":false,
  "chimeModeActive":false,
  "incs":[]
}
```


15.1 *Status Retrieval*

Request

Verb	*GET*
Query Parameters	(none)

Response (Success)

Return Code	200
Content	Area object structure

Response (Error)

See standard errors defined in OII Base Specification.

15.2 *Arm*

This command initiates the arming of an area. In case the access rights of the user allow arming and the area is ready to arm, the command will be accepted. Once the arming has been done, the status of the area will change accordingly (i.e. "armed":true).

In rare cases it may happen that the area will not successfully arm, although the command has been accepted. Thus a client shall not assume that the arming will necessarily be successful even though the request accepted.

Request

Verb	*POST*
Query Parameters	(none)

Request Body:

Name	Type	Value Range/Format	Description
@cmd	String	ARM	Indicates arming activity to be started
bypassOffNormalDevices	Bool	true/false	Bypass all devices that are off normal before arming.
exitDelay	String	ZERO USERDEFAULT EXTENDED	Defines whether the arming should happen without a delay (zero) with the user configured default exit delay or with the extended exit delay as configured for the area.

Response (Success)

Return Code	202
Content	-

Response (Error)

Return Code	Content	Description
409 (Conflict)	Area not ready to arm	Response if arm command requested on an area which is not ready to arm
409 (Conflict)	RPS connected to Panel	Response to an arming attempt when RPS is connected to the Panel which prevents the area from being armed
409 (Conflict)	Area not user armable	Response if arm command requested on an area which cannot be armed since it is not OII Armable

Additionally see standard errors defined in OII Base Specification.

15.3 *Disarm*

Request

Verb	*POST*
Query Parameters	(none)

Request Body:

Name	Type	Value Range/Format	Description
@cmd	String	DISARM	

Response (Success)

Return Code	202
Content	-

Response (Error)

Return Code	Content	Description
409 (Conflict)	Area not ready to disarm	Response if disarm command requested on an area which is not ready to disarm
409 (Conflict)	Area not user disarmable	Response if disarm command requested on an area which cannot be disarmed since it is not OII Armable

Additionally see standard errors defined in OII Base Specification.

15.4 *Start Walktest*

Starts a walktest for an area. For each walktest a new WalktestResource is created. A reference to the currently running walktest is provided in the area object structure. Only one walktest can be active in an area at a given point in time. The walktest can be stopped using the stop walktest operation on the area or by deleting the corresponding walktest resource.

When the walktest is started, all devices contained in the area will switch to walktest mode. The individual device provides the information whether the walktest is successful.

Request

Verb	*POST*
Query Parameters	(none)

Request Body:

Name	Type	Value Range/Format	Description
@cmd	String	STARTWALKTEST	
includedPoints	String	ALL SEISMICONLY	Specifies which points should be put into test mode.

Response Start Walktest (Success)

Return Code	200
Content	JSON Object containing a single key “walktest” with the walktest URLs as a String value e.g. {“walktest”:”/walktest/1”}

Response Start Walktest (Error)

Trying to start a walktest in an area which is currently running a walktest is rejected with a 409 return code and the defined String content (not JSON). The currently running walktest remains unchanged and running. In case a client wants to restart a walktest, it needs to first stop the walktest and then start a new one.

Return Code	Content	Description
409 (Conflict)	Walktest already running	Response if area is already running a walktest
409 (Conflict)	No points to walktest	Response if there are no points to be walktested in the area
409 (Conflict)	Area armed	Response if the area cannot be walktested because the area is armed

15.5 *Stop Walktest*

This command is used to stop the walktest.

Request

Verb	*POST*
Query Parameters	(none)

Request Body:

Name	Type	Value Range/Format	Description
@cmd	String	STOPWALKTEST	

Response (Success)

Return Code	202
Content	-

Response (Error)

See standard errors defined in OII Base Specification.

15.6 *Start Motion Detector Test*

This operation is used to start motion detector test in the area. A motion detector tests puts all motion detectors configured in that area into a test mode. The devices under motion detector test will not report any status changes to the panel and the device will locally indicate a motion being detected. If area running motion detector tests is armed, motion detector tests will be stopped automatically.

Request

Verb	*POST*
Query Parameters	(none)

Request Body:

Name	Type	Value Range/Format	Description
@cmd	String	STARTMDTEST	

Response (Success)

Return Code	202
Content	-

Response (Error)

See standard errors defined in OII Base Specification.

15.7 *Stop Motion Detector Test*

This operation stops a motion detector test running in the area.

Request

Verb	*POST*
Query Parameters	(none)

Request Body:

Name	Type	Value Range/Format	Description
@cmd	String	STOPMDTEST	

Response (Success)

Return Code	202
Content	-

Response (Error)

See standard errors defined in OII Base Specification.

15.8 *Start Chime Mode*

When set to chime mode, all keypads will chime, whenever a point configured for chime goes off normal. This is usually applied in scenarios like shops, where a chime starts whenever a customer enters or leaves the shop.

Request

Verb	*POST*
Query Parameters	(none)

Request Body:

Name	Type	Value Range/Format	Description
@cmd	String	STARTCHIMEMODE	

Response (Success)

Return Code	202
Content	-

Response (Error)

Return Code	Content	Description
409 (Conflict)	Not supported by Control Panel Area	Response from area when Chime mode attempted to be turned on in Control Panel Area

Additionally see standard errors defined in OII Base Specification.

15.9 *Stop Chime Mode*

This operation stops chime mode running in an area.

Request

Verb	*POST*
Query Parameters	(none)

Request Body:

Name	Type	Value Range/Format	Description
@cmd	String	STOPCHIMEMODE	

Response (Success)

Return Code	202
Content	-

Response (Error)

Return Code	Content	Description
409 (Conflict)	Not supported by Control Panel Area	Response from area when Chime mode attempted to be stopped in Control Panel Area

Additionally see standard errors defined in OII Base Specification.

15.10 *Start Bell Test*

This operation activates all outputs for a defined time span which are configured to be included in a bell test. Configuration is done via RPS. Bell test cannot be stopped over OII. If an area is armed, only indicators on the keypad configured in the area will be activated.

Request

Verb	*POST*
Query Parameters	(none)

Request Body:

Name	Type	Value Range/Format	Description
@cmd	String	BELLTEST	

Response (Success)

Return Code	202
Content	-

Response (Error)

See standard errors defined in OII Base Specification.

15.11 *ArmingInfo*

This command provides information about the reasons why the area may not be ready to arm or disarm. Furthermore, it indicates whether the area is ready to force arm. If the area is queried when the area is already ready to arm or ready to disarm, then the reasons arrays will be empty.

Request

Verb	*POST*
Query Parameters	(none)

Request Body:

Name	Type	Value Range/Format	Description
@cmd	String	ARMINGINFO	

Response (Success)

Return Code	200
Content	ArmingInfo Object Structure

Object Structure

Name	Type	Value Range/Format	Description
armed	boolean	true/false	Indicates whether area is armed
readyToArm	boolean	true/false	Indicates whether is area is ready to arm. If the area is already armed, then this flag will be false
readyToForceArm	boolean	true/false	Indicates whether this area can be armed by bypassing off normal devices. If the area is already armed, this flag will be false.
readyToDisarm	boolean	true/false	Indicates whether this area can be disarmed. Will be false if the area is already disarmed.

Name		Type	Value Range/Format	Description
whyNotReadyToArm		object	-	Nested objects providing reasons why the area is not ready to arm (the objects contained within are listed below)
	bypassableFaultedDevices	Array	Array of urls	List of off normal devices that can be bypassed while arming
whyNotReadyToForceArm		object	-	Nested objects providing reasons why the area cannot be armed by bypassing off normal devices (the objects contained within are listed below)
	nonBypassableFaultedDevices	Array	Array of urls	List of off normal devices that cannot be bypassed
	tooManyFaultedDevices	Array	Array of urls	List of all off normal devices, the count of which exceeds the number of devices that can be bypassed to arm the area
	tooManyBypassedDevices	bool	true/false	Indicates that the area cannot be force armed because there are too many bypassed devices

Name	Type	Value Range/Format	Description
devicesBypassedTooManyTimes	Array	Array of urls	List of all off normal devices which have been bypassed multiple times exceeded the number of times a device can be bypassed
systemWideDevicesFaulted	Array	Array of urls	List of all off normal devices which are considered as system wide devices
relatedAreasNotReadyToArm	Array	Array of urls	List of related areas which are not ready to arm which make this area not ready to arm
relatedAreasMustBeArmedFirst	Array	Array of urls	List of related areas which need to be armed first before this area can be armed
pendingIncidentsInArea	boolean	true/false	Flag indicates whether there is an unhandled incident in the area
areaInWalktest	boolean	true/false	Flag indicates whether the area is in walktest
whyNotReadyToDisarm	object	-	Nested objects providing reasons why the area is not ready to disarm (The objects contained within are listed below)

Name		Type	Value Range/Format	Description
	relatedAreasNotReadyToDisarm	Array	Array of urls	List of related areas which are not ready to disarm which make this area not ready to disarm
	relatedAreasMustBeDisarmedFirst	Array	Array of urls	List of related areas which need to be disarmed before this area can be disarmed
	virtualOutputsOffPreventsDisarming	Array	Array of urls	List of virtual outputs which are off preventing the area from disarming
	activeBlockingTimePreventsDisarming	boolean	true/false	Flag indicates whether a blocking time is active which prevents the area from disarming

Response (Error)

If the panel is processing an arming info request already when there is another request, the panel will return a 503 (Service Unavailable) response code. A client can try again at a later point of time.

Example

```
{
  "armed": false,
  "readyToArm": false,
  "readyToForceArm": false,
  "readyToDisarm": false,
  "whyNotReadyToArm": {
    "bypassableFaultedDevices": [ "/1.1.point.1.1",
      "/1.1.point.2.2" ]
  },
  "whyNotReadyToForceArm": {
    "nonBypassableFaultedDevices": [ "/1.1.point.2.3",
      "/1.1.point.2.4" ],
    "tooManyFaultedDevices": false
    "devicesBypassedTooManyTimes": [ "/1.1.point.4.4" ],
    "systemWideDevicesFaulted": [ "/1.1.point.4.5" ],
    "relatedAreasNotReadyToArm": [ "/1.1.area.2.2" ],
    "relatedAreasMustBeArmedFirst": [ "/1.1.area.2.3" ],
    "pendingIncidentsInArea": true,
    "areaInWalktest": true
  },
  "whyNotReadyToDisarm": {
    "relatedAreasNotReadyToDisarm": ["/1.1.area.2.6",
     "/1.1.area.2.7"],
    "relatedAreasMustBeDisarmedFirst": ["/1.1.area.2.5"],
    "virtualOutputsOffPreventsDisarming": ["/1.1.Output.10.1"],
    "activeBlockingTimePreventsDisarming": false
  }
}
```


16 AreaList

The AreaList resource wraps a list of areas, to provide status of and access to multiple areas at the same time.

Overview

OII Type	Supported Operations
IN.areaList.1, OII.list.1	Status Retrieval Arm Disarm Start Walktest Stop Walktest Start Motion Detector Test Mode Stop Motion Detector Test Mode Start Bell Test Start Chime Mode Stop Chime Mode

Object Structure

Name	Type	Value Range/Format	Description
@type	Arrays of strings	[IN.areaList.1, OII.list.1]	Fixed type identifier
@self	String	URL starting with "/"	Link to the current resource
List	Array	Array of Area Objects of type IN.area.1	

Example

```
{
  "@type": ["IN.areaList.1","OII.list.1"],
  "@self": "/areas",
  "list": [
    {
      "@type": ["IN.area.1"],
      "@self": "/1.1.Area.1.1",
      "armed":false,
      "transitionalState":"",
      "readyToArm":true,
      "readyToDisarm":false,
      "oiiArmable":true,
      "numberOfBypassedDevices":2,
      "walkTestActive": false,
      "motionDetectorTestActive":false,
      "chimeModeActive":false,
      "incidents":[]
    },
    {
      "@type": ["IN.area.1"],
      "@self": "/1.1.Area.2.1",
      "armed":true,
      "transitionalState":"",
      "readyToArm":false,
      "readyToDisarm":true,
      "oiiArmable":false,
      "numberOfBypassedDevices":0,
      "walkTestActive": false,
      "motionDetectorTestActive":false,
      "chimeModeActive":false
      "incidents":[]
    }
  ]
}
```


16.1 *Status Retrieval*

The area list allows retrieval of status information of multiple areas. The areas that are to be fetched are selected using query parameters (url). If no parameters are specified, the status of all areas is provided.

Verb	*GET*		
Query Parameters	url	Resource ID as specified for OII.list.1	optional

Response (Success)

Return Code	200
Content	AreaList object structure. The list object will only contain resources which match the defined filters given in the query parameters.

Response (Error)

See standard errors defined in OII Base Specification.

16.2 *Arm*

The AreaList allows arming and disarming of multiple areas with a single request. The areas that are to be armed or disarmed are selected using query parameters (url). Thus, the client can use the AreaList resource to arm/disarm one, multiple or all areas in the panel.

In case some of the areas selected by the filter are already armed, the arm request will be accepted and the remaining areas will be armed. In case some of the areas selected by the filter are not ready to arm, the behaviour would be decided by the presence of the atomic flag in the query.

Request

Verb	POST		
Query Parameters	url	Resource ID as specified for OII.list.1	optional
	atomic	If set, command is only executed when execution on selected resources is possible.	optional

Request Body:

Same as for IN.area.1

Response (Success)

Return Code	202
Content	-

Response (Error)

Return Code	Content	Description
409 (Conflict)	Comma separated list of urls	List of resource urls which prevented the command from being executed when atomic is true
409 (Conflict)	Not executed	String describing that application rules prevented execution of the command on any element of the list. Will be only retrurned when atomic is false.

16.3 *Disarm*

The command disarms the selected areas. In case some of the selected areas are already disarmed, the remaining areas will be disarmed. If a few areas are not ready to disarm, the behaviour would be decided by the presence of the atomic flag in the query.

Request

Verb	POST		
Query Parameters	url	Resource ID as specified for OII.list.1	optional
	atomic	If set, command is only executed when execution on selected resources is possible.	optional

Request Body:

Same as for IN.area.1

Response (Success)

Return Code	202
Content	-

Response (Error)

Return Code	Content	Description
409 (Conflict)	Comma separated list of urls	List of resource urls which prevented the command from being executed when atomic is true

Return Code	Content	Description
409 (Conflict)	Not executed	String describing that application rules prevented execution of the command on any element of the list. Will be only returned when atomic is false.

16.4 *Start Walktest*

Similar as with arming and disarming, the AreaList resource can be used to start and stop a walktest in multiple areas, which are selected using query parameters (url).
All selected areas will be combined to a single walktest. A Walktest resource is created for this walktest. The link to the walktest is provided as a response to the start walktest request. While a walktest is running, no areas can be added or removed from the waltkest. But a new walktest can be started with areas that are currently not part of a walktest.

The walktest can be stopped in three different ways:

1. Send a stop walktest request to the area list specifying atleast one area that is contained in the walk test
2. Send a stop walktest request to an individual area which is part of the walktest
3. Send a stop walktest request to the walktest list resource
4. Send a stop walktest request to the individual walktest resource

The different possibilities have been introduced to comply with legacy systems.

Request

Verb	POST		
Query Parameters	url	Resource ID as specified for OII.list.1	optional
	atomic	If set, command is only executed when execution on selected resources is possible.	optional

Request Body:

Same as for IN.area.1

Response (Success)

Same as for IN.area.1

If atomic flag is not set, walktest will be started in areas which can be walktested and a response of 200 will be provided. Clients will need to check the walktest resource to confirm which areas are actually included into the walktest.

Response (Error)

Return Code	Content	Description
409 (Conflict)	Comma separated list of urls	List of resource urls which prevented the command from being executed when atomic is true
409 (Conflict)	Not executed	String describing that application rules prevented execution of the command on any element of the list. Will be only retrurned when atomic is false.

16.5 *Stop Walktest*

This command stops all walktests in the selected areas. In case some areas do not have a walktest running, the walktest in the remaining areas is stopped and success is reported.

Request

Verb	POST		
Query Parameters	url	Resource ID as specified for OII.list.1	optional
	atomic	If set, command is only executed when execution on selected resources is possible.	optional

Request Body:

Same as for IN.area.1

Response (Success)

Same as for IN.area.1

Response (Error)

Same as for IN.area.1

16.6 *Start Motion Detector Test*

Similar as with arming and disarming, the AreaList resource can be used to start and stop a motion detector test in multiple areas, which are selected using query parameters (url).

A motion detector test puts all motion detectors configured in that area into a test mode. The devices under motion detector test will not report any status changes to the panel and the device will locally indicate a motion being detected. If area running motion detector tests is armed, motion detector tests will be stopped automatically.

Request

Verb	POST		
Query Parameters	url	Resource ID as specified for OII.list.1	optional
	atomic	If set, command is only executed when execution on selected resources is possible.	optional

Request Body:

Same as for IN.area.1

Response (Success)

Return Code	202
Content	-

Response (Error)

Return Code	Content	Description
409 (Conflict)	Comma separated list of urls	List of resource urls which prevented the command from being executed when atomic is true
409 (Conflict)	Not executed	String describing that application rules prevented execution of the command on any element of the list. Will be only retrurned when atomic is false.

16.7 *Stop Motion Detector Test*

Request

Verb	POST		
Query Parameters	url	Resource ID as specified for OII.list.1	optional
	atomic	If set, command is only executed when execution on selected resources is possible.	optional

Request Body:

Same as for IN.area.1

Response (Success)

Return Code	202
Content	-

Response (Error)

Same as for IN.area.1

16.8 *Start Chime Mode*

Similar as with arming and disarming, the AreaList resource can be used to start and stop the Chime Mode in selected areas.

When set to chime mode, all keypads will chime, whenever a point configured for chime goes off normal. This is usually applied in scenarios like shops, where a chime starts whenever a customer enters or leaves the shop.

Request

Verb	POST		
Query Parameters	url	Resource ID as specified for OII.list.1	optional
	atomic	If set, command is only executed when execution on selected resources is possible.	optional

Request Body:

Same as for IN.area.1

Response (Success)

Return Code	202
Content	-

Response (Error)

Return Code	Content	Description
409 (Conflict)	Comma separated list of urls	List of resource urls which prevented the command from being executed when atomic is true
409 (Conflict)	Not executed	String describing that application rules prevented execution of the command on any element of the list. Will be only retrurned when atomic is false.

16.9 *Stop Chime Mode*

Request

Verb	POST		
Query Parameters	url	Resource ID as specified for OII.list.1	optional
	atomic	If set, command is only executed when execution on selected resources is possible.	optional

Request Body:

Same as for IN.area.1

Response (Success)

Return Code	202
Content	-

Response (Error)

Same as for IN.area.1

16.10 *Bell Test*

This operation activates all outputs for the selected areas. The specific bell test behaviour is area specific. Configuration of the bell test behaviour is done via RPS.

Request

Verb	POST		
Query Parameters	url	Resource ID as specified for OII.list.1	optional
	atomic	If set, command is only executed when execution on selected resources is possible.	optional

Request Body:

Same as for IN.area.1

Response (Success)

Return Code	202
Content	-

Response (Error)

Return Code	Content	Description
409 (Conflict)	Comma separated list of urls	List of resource urls which prevented the command from being executed when atomic is true
409 (Conflict)	Not executed	String describing that application rules prevented execution of the command on any element of the list. Will be only retrurned when atomic is false.

17 Walktest

The walktest resource represents a walktest that contains one or multiple areas. A walktest resource is created during runtime when a client or a user starts a walktest. The walktest contains information which areas should participate in the walktest and if the walktest has already stared in that area. Additionally, it allows stopping the walktest. Walktests are started via the area or area list resource. Each walktest is referenced by a walktest list.

Overview

OII Type	Supported Operations
IN.walktest.1	Status Retrieval Stop Walktest

Object Structure

Name	Type	Value Range/Format	Description
@type	Array of strings	["IN.walktest.1"]	Fixed type identifier
@self	String	URL starting with "/"	Link to the current resource
user	String		Id of the user that started the walktest (e.g. “123”)
interface	String	OII or KEYPAD	Interface that was used to start the keypad. Possible values are KEYPAD and OII.
wt	Array of String, String	List of tuples of “area” and “wtStatus”	• “area”: URL to the area that is part of this walktest • "wtStatus”: Walktest status of the area. Possible values are PENDING (walktest requested but not started), STARTED, FAILED (walktest requested but area did not start test, e.g. because of area being armed)

17.1 *Status Retrieval*

Request

Verb	*GET*
Query Parameters	(none)

Response (Success)

Return Code	200
Content	Walktest object structure

Response (Error)

See standard errors defined in OII Base Specification.

Example


```
{
  "@type": [
    "IN.walktest.1"
  ],
  "@self": "/walktest/274F1EC2086C40AB8601DF998431DA46 ",
  "user": "123",
  "interface": "OII",
  "wt": [
    {
      "area": "/1.1.Area.2.4",
      "wtStatus": "PENDING"
    },
    {
      "area": "/1.1.Area.2.1",
      "wtStatus": "STARTED"
    },
    {
      "area": "/1.1.Area.2.36",
      "wtStatus": "FAILED"
    }
  ]
}
```


17.2 *Stop Walktest*

This operation stops the walktest in all related areas and points that are part of this test

Request

Verb	*DELETE*
Query Parameters	(none)

Request Body:

-

Response (Success)

Return Code	202
Content	-

Response (Error)

See standard errors defined in OII Base Specification.

18 Walktest List

The walktest list resource allows access to all walktests currently running in the panel. The walktest list will not create any events.

Overview

OII Type	Supported Operations
IN.walktestList.1, OII.list.1	Status Retrieval Stop Walktest

Object Structure

Name	Type	Value Range/Format	Description
@type	Array of strings	[IN.walktestList.1, OII.list.1]	Fixed type identifier
@self	String	URL starting with "/"	Link to the current resource
list	Array	Array of Walktest Objects of type IN.walktest.1	

18.1 *Status Retrieval*

Request

Verb	*GET*		
Query Parameters	url	Resource ID as specified for OII.list.1	optional

Response (Success)

Return Code	200
Content	WalktestList object structure. The list object will only contain resources which match the defined filters given in the query parameters.

Response (Error)

See standard errors defined in OII Base Specification.

18.2 *Stop Walktest*

This operation stops all walktests that are specified using the list filter.

Request

Verb	*DELETE*		
Query Parameters	url	Resource ID as specified for OII.list.1	optional
	atomic	If set, command is only executed when execution on selected resources is possible.	optional

Request Body:

-

Response (Success)

Return Code	202
Content	-

Response (Error)

See standard errors defined in OII Base Specification.

19 Internal Program

The internal program resource allows investigating the status of internal programs configured in the map and allows activating or deactivating the internal program. In case an alarm occurs, the alarm will be referenced by the area in which it occurred not at the internal program. Similarly, the incident will not indicate the internal program but the area that detected the alarm.

Overview

OII Type	Supported Operations
IN.internalProgram.1	Status Retrieval Activate Internal Program Deactivate Internal Program IP Arming Info

Object Structure

Name	Type	Value Range/Format	Description
@type	Array of strings	["IN.internalProgram.1"]	Fixed type identifier
@self	String	URL starting with "/"	Link to the current resource
active	boolean		True, if the internal program is active. False, otherwise.

19.1 *Status Retrieval*

Request

Verb	*GET*
Query Parameters	(none)

Response (Success)

Return Code	200
Content	Internal program object structure

Response (Error)

See standard errors defined in OII Base Specification.

19.2 *Activate Internal Program*

This operation activates internal program in the panel.

Request

Verb	*POST*
Query Parameters	(none)

Request Body:

Name	Type	Value Range/Format	Description
@cmd	String	ACTIVATE	

Response (Success)

Return Code	202
Content	-

Response (Error)

In case the internal program could not be activated due to some related devices being active, the request is going to be rejected.

Return Code	Content	Description
409 (Conflict)	Devices off-normal	Returned when internal program could not be activated since at least one device is off normal which prevents it from being activated.

19.3 *Deactivate Internal Program*

This operation deactivates internal program in the panel.

Request

Verb	*POST*
Query Parameters	(none)

Request Body:

Name	Type	Value Range/Format	Description
@cmd	String	DEACTIVATE	

Response (Success)

Return Code	202
Content	-

Response (Error)

See standard errors defined in OII Base Specification.

19.4 *IP Arming Info*

This command provides information about the reasons why the internal program cannot be activated. It provides the list of related device urls that prevent the internal program from being activated.

Request

Verb	*POST*
Query Parameters	(none)

Request Body:

Name	Type	Value Range/Format	Description
@cmd	String	IPARMINGINFO	

Response (Success)

Return Code	200
Content	IPArmingInfo Object Structure

Object Structure

Name	Type	Value Range/Format	Description
activated	boolean	true/false	Indicates whether internal program is activated
canBeActivated	boolean	true/false	Indicates whether is internal program can be activated. If it is already activated, then this flag will be false
deviceList	Array of urls		The list of devices urls that prevent the internal program from being activated

20 Internal Program List

This resource allows batched access to internal programs. The maximum length of the list is 14, as the MAP supports up to 14 internal programs

Overview

OII Type	Supported Operations
IN.internalProgramList.1, OII.list.1	Status Retrieval Activate Internal Program Deactivate Internal Program

Object Structure

Name	Type	Value Range/Format	Description
@type	Array of strings	[IN.internalProgramList.1, OII.list.1]	Fixed type identifier
@self	String	URL starting with "/"	Link to the current resource
list	Array	Array of Internal Program Objects of type IN.internalProgram.1	

20.1 *Status Retrieval*

Request

Verb	GET		
Query Parameters	url	Resource ID as specified for OII.list.1	optional

Response (Success)

Return Code	200
Content	Internal Program List object structure. The list object will only contain resources which match the defined filters given in the query parameters.

Response (Error)

See standard errors defined in OII Base Specification.

20.2 *Activate Internal Program*

This operation activates or deactivates all internal programs specified by the filter.

Request

Verb	POST		
Query Parameters	url	Resource ID as specified for OII.list.1	optional
	atomic	If set, command is only executed when execution on selected resources is possible.	optional

Request Body:

Same as IN.internalProgram.1

Response (Success)

Same as IN.internalProgram.1

Response (Error)

Return Code	Content	Description
409 (Conflict)	Comma separated list of urls	List of resource urls which prevented the command from being executed when atomic is true

Return Code	Content	Description
409 (Conflict)	Not executed	String describing that application rules prevented execution of the command on any element of the list. Will be only retrurned when atomic is false.

20.3 *Deactivate Internal Program*

Request

Verb	*POST*		
Query Parameters	url	Resource ID as specified for OII.list.1	optional
	atomic	If set, command is only executed when execution on selected resources is possible.	optional

Request Body:

Same as IN.internalProgram.1

Response (Success)

Same as IN.internalProgram.1

Response (Error)

Same as IN.internalProgram.1

21 Devices

The following section describes the resource types and resource models for all devices supported by MAP.

21.1 *IN.dev.1*

This resource type is the basic resource type for all MAP devices (except for a few specific devices). It thus includes the commonly available properties and commands for devices in the MAP system.

Overview

OII Type	Supported Operations
IN.dev.1	Status Retrieval

Object Structure

Name	Type	Value Range/Format	Description
@type	Array of strings	[“IN.dev.1”]	Fixed type identifier
@self	String	URL starting with “/”	Link to the current resource
opState	String	"OK"/"INIT"/"PROGRAMMING"/"MALFUNCTION"	The operational state of the device shows the general condition the device is in. It also specifically identifies whether the device is working properly at the moment, so that all status information is updated and commands are processed. Potential Values are: • OK: Device is running properly • INIT: Device is initializing. This should only be the case during start up. • PROGRAMMING: The firmware or configuration of the device is being updated • MALFUNCTION: The device is not working properly at the moment. Any state information provided by the resource should be considered invalid. Details to the condition of the device can be taken from the incidents of the device.

Name	Type	Value Range/Format	Description
incs	Array	Array of links to currently pending incidents related to this device	A list of incidents that relate to this device. In case the opState is MALFUNCTION the incident will give more detailed information about the error condition the device is in.

21.1.1 Status Retrieval

Request

Verb	*GET*
Query Parameters	(none)

Response (Success)

Return Code	200
Content	Device object structure

Response (Error)

See standard errors defined in OII Base Specification.

21.2 *Device lists*

To allow batch operations on devices of the same type, MAP supports list resources for each of the defined device types.

Currently these are the device list resources and the operations supported by them –

List Resource	Description	Operations
IN.deviceList.1	List resource for all the devices	Status Retrieval
IN.supervConnList.1	List resource for IN.supervConn.1 types of resources	Status Retrieval Disable/Enable
IN.keypadList.1	List resource for IN.keypad.1 types of resources	Status Retrieval Disable/Enable Activate/Deactivate
IN. lsnGwList.1	List resource for IN. lsnGateway.1 types of resources	Status Retrieval Disable/Enable

List Resource	Description	Operations
IN.powerSupList.1	List resource for IN. powerSupply.1 types of resources	Status Retrieval Disable/Enable
IN.blocklockList.1	List resource for IN. blocklock.1 types of resources	Status Retrieval Disable/Enable
IN.smartkeyList.1	List resource for IN. powerSupply.1 types of resources	Status Retrieval Disable/Enable
IN.outputList.1	List resource for IN. output.1 types of resources	Status Retrieval Disable/Enable Turn On/Off
IN.pointList.1	List resource for IN. point.1 types of resources	Status Retrieval Disable/Enable Bypass/Unbypass
IN.fireDetList.1	List resource for IN. fireDetector.1 types of resources	Status Retrieval Disable/Enable Bypass/Unbypass
IN.keyswitchList.1	List resource for IN. keyswitch.1 types of resources	Status Retrieval Disable/Enable
IN.couplerList.1	List resource for IN. coupler.1 types of resources	Status Retrieval Disable/Enable
IN.lsnBusList.1	List resource for IN. lsnBus.1 types of resources	Status Retrieval Disable/Enable
IN.lsnAuxList.1	List resource for IN. lsnAux.1 types of resources	Status Retrieval Disable/Enable
IN.batChgList.1	List resource for IN. batterCharger.1 types of resources	Status Retrieval Disable/Enable
IN.batteryList.1	List resource for IN. battery.1 types of resources	Status Retrieval Disable/Enable
IN.mainsList.1	List resource for IN. mains.1 types of resources	Status Retrieval Disable/Enable
IN.gndFltList.1	List resource for IN. groundFault.1 types of resources	Status Retrieval Disable/Enable
IN.psCanOpList.1	List of resource for IN.psCanOp.1 types of resources	Status Retrieval
IN.powerDevList.1	List resource for IN. powerDevice.1 types of resources	Status Retrieval Disable/Enable

The request body for each of the operations is the same as that for the individual resources explained in the following sections.

Success responses are the same as specified for the individual resources.

In case execution of commands in any device list fails, the following errors will be given based on the value of the atomic flag.

Return Code	Content	Description
409 (Conflict)	Comma separated list of urls	List of resource urls which prevented the command from being executed when atomic is true
409 (Conflict)	Not executed	String describing that application rules prevented execution of the command on any element of the list. Will be only returned when atomic is false.

21.3 *Disable*

The resource type IN.dsbl.1 is used to indicate for a resource that it can be disabled. The resource type has only one property and two commands. There will be no resource that only has the IN.dsbl.1 type alone. Thus, the resource type is rather an interface that is additionally used by existing resource types like IN.point.1 to indicate that disabling is possible and supported.

Name	Type	Value Range/Format	Description
enabled	Bool	True/False	True if device is currently enabled, otherwise false

21.3.1 Disable

This command initiates the disabling of a device. As long as the device is disabled, MAP shall ignore status changes from the device. Once disabled, the status of the device will change accordingly (i.e. “enabled”: true).

A device is usually disabled when it is considered to be malfunctioning.

Request

Verb	*POST*
Query Parameters	(none)

Request Body:

Name	Type	Value Range/Format	Description
@cmd	String	DISABLE	Indicates initiating disabling of the device

Response (Success)

Return Code	202
Content	-

Response (Error)

Return Code	Content	Description
409 (Conflict)	Area Armed	Response if the area containing the device is armed
409 (Conflict)	Area not configured for device	Response if the device is not configured in any area
409 (Conflict)	Keypad deactivated	Response if disable is tried to be executed on a deactivated keypad

Additionally see standard errors defined in OII Base Specification.

21.3.2 Enable

Request

Verb	*POST*
Query Parameters	(none)

Request Body:

Name	Type	Value Range/Format	Description
@cmd	String	ENABLE	Indicates initiating enabling of the device

Response (Success)

Return Code	202
Content	-

Response (Error)

Return Code	Content	Description
409 (Conflict)	Area Armed	Response if the area containing the device is armed
409 (Conflict)	Area not configured for device	Response if the device is not configured in any area

Additionally see standard errors defined in OII Base Specification.

21.4 *Bypass*

The resource type IN.byp.1 is used to indicate for a resource that it can be bypassed. The resource type has only one property and two commands. There will be no resource that only has the IN.byp.1 type alone. Thus, the resource type is rather an interface that is additionally used by existing resource types like IN.point.1 to indicate that bypassing is possible and supported.

Name	Type	Value Range/Format	Description
bypassed	Bool	True/False	True if device is currently bypassed, otherwise false

21.4.1 Bypass

Bypassing a device will mean that only non bypassable incidents will be generated from the device. Usually used to allow arming of an area even if the device is not normal e.g. an open window.

Request

Verb	*POST*
Query Parameters	(none)

Request Body:

Name	Type	Value Range/Format	Description
@cmd	String	BYPASS	Indicates initiating bypass of the device

Response (Success)

Return Code	202
Content	-

Response (Error)

Return Code	Content	Description
409 (Conflict)	Area Armed	Response if the area containing the device is armed
409 (Conflict)	Area not configured for device	Response if the device is not configured in any area
409 (Conflict)	Device in walktest	Response if the device is currently in walktest

Note: It is configurable (through MAP RPS) as to whether bypass should be allowed when the area is armed.

Additionally see standard errors defined in OII Base Specification.

21.4.2 Unbypass

Request

Verb	*POST*
Query Parameters	(none)

Request Body:

Name	Type	Value Range/Format	Description
@cmd	String	UNBYPASS	Indicates initiating unbypass of the device

Response (Success)

Return Code	202
Content	-

Response (Error)

Return Code	Content	Description
409 (Conflict)	Area Armed	Response if the area containing the device is armed
409 (Conflict)	Area not configured for device	Response if the device is not configured in any area

Note: It is configurable (through MAP RPS) as to whether unbypass should be allowed when the area is armed.

Additionally see standard errors defined in OII Base Specification.

21.5 *Walktest*

The resource type IN.walkT.1 is used to indicate for a resource that it can be walktested. The resource type has only one property and one command. There will be no resource that only has the IN.walkT.1 type. Thus, the resource type is rather an interface that is additionally used by existing resource types like IN.point.1 to indicate that walktest is supported.

Name	Type	Value Range/Format	Description
walktest	String	"NONE"/"STARTED"/"SUCCESS"	Gives information on the walktest status. Possible values are: NONE: if no walktest is running for the device. STARTED: if walktest for the device started but device has not been successfully walktested. SUCCESS: a walktest is running

Name	Type	Value Range/Format	Description
			and this device has been successfully tested in the scope of this walktest.

21.5.1 DIAGNOSE

A walktestable device provides some diagnostic information that can be fetched using the diagnose command. (This information has not been included as a property of the resource as it is changing only rarely and is usually only used in preparation of a walktest. Thus, moving it into a command allows to save bandwidth in most scenarios.)

Request

Verb	*POST*
Query Parameters	(none)

Request Body:

Name	Type	Value Range/Format	Description
@cmd	String	DIAGNOSE	Retrieves diagnostic information of the device

Response (Success)

Return Code	200
Content	JSON structure with information on Days since last activity, Last Walktest, Walktest advised

Response Structure

Name	Type	Value Range/Format	Description
daysSinceLastActivity	Number	0 - 255	Number of days since the last activity from the device

Name	Type	Value Range/Format	Description
lastWalktestDate	String	2014-07-20T14:11:42+05:30	Time and date (in OII base specification compliant format)
walktestRecommended	Bool	True/False	True if a walktest is recommended for the device, else false

Response (Error)

See standard errors defined in OII Base Specification.

21.6 *DE Module*

The resource type IN.deModule.1 extends the IN.dev.1 resource type with an additional command of firmware version. It models the DE Module. It can be disabled and implements the IN.dsbl.1 interface. The resource structure will contain the attributes of IN.dev.1 and IN.dsbl.1.
It cannot be bypassed or walktested.

Overview

OII Type	Supported Operations
IN.dev.1 IN.dsbl.1 IN.deModule.1	Status Retrieval Enable Disable Firmware Version

21.6.1 Firmware Version

The firmware version command retrieves the version of the firmware running on the DE Module.

Request

Verb	*POST*
Query Parameters	(none)

Request Body

Name	Type	Value Range/Format	Description
@cmd	String	FWVERSION	

Response (Success)

Return Code	200
Content	Firmware Version Object Structure

Object Structure

Name	Type	Value Range/Format	Description
firmwareVersion	String	“1.0.1”	Firmware version of the DEModule in format MAJOR.MINOR.MICRO

Response (Error)

If the panel is processing a firmware version request already when there is another request, the panel will return a 503 (Service Unavailable) response code. A client can try again at a later point of time.

21.7 *Power Supply*

The resource type IN.powerSupply.1 extends the IN.dev.1 resource type with an additional command of firmware version. It models the Power Supply. It can be disabled and implements the IN.dsbl.1 interface. The resource structure will contain the attributes of IN.dev.1 and IN.dsbl.1. This resource type cannot be bypassed or walktested.

Overview

OII Type	Supported Operations
IN.dev.1 IN.dsbl.1 IN.powerSupply.1	Status Retrieval Enable Disable Firmware Version

21.7.1 Firmware Version

The firmware version command retrieves the version of the firmware running on the Power Supply.

Request

Verb	*POST*
Query Parameters	(none)

Request Body

Name	Type	Value Range/Format	Description
@cmd	String	FWVERSION	

Response (Success)

Return Code	200
Content	Firmware Version Object Structure

Object Structure

Name	Type	Value Range/Format	Description
firmwareVersion	String	“1.0.1”	Firmware version of the power supply in format MAJOR.MINOR.MICRO

Response (Error)

If the panel is processing a firmware version request already when there is another request, the panel will return a 503 (Service Unavailable) response code. A client can try again at a later point of time.

21.8 *AC Mains*

The resource type IN.mains.1 models the ac input of the power supply. It provides the same interface as the resource type IN.dev.1. It can be disabled and implements the IN.dsbl.1 interface. It can be bypassed and implements the IN.byp.1 interface. The resource structure will contain attributes of IN.dev.1, IN.dsbl.1 and IN.byp.1. This resource type cannot be walktested.

OII Type	Supported Operations
IN.dev.1 IN.dsbl.1 IN.byp.1 IN. mains.1	Status Retrieval Enable Disable Bypass Unbypass

21.9 *Battery*

The resource type IN.battery.1 models the battery input of the power supply. It provides the same interface as the resource type IN.dev.1. It can be disabled and implements the IN.dsbl.1 interface. It can be bypassed and implements the IN.byp.1 interface. This resource type cannot be walktested.

Overview

OII Type	Supported Operations
IN.dev.1 IN.dsbl.1 IN.byp.1 IN. battery.1	Status Retrieval Enable Disable Bypass Unbypass

21.10 *Battery Charger*

The resource type IN. batteryCharger.1 models the battery charger of the power supply. It provides the same interface as the resource type IN.dev.1. It can be disabled and implements the IN.dsbl.1 interface. It can be bypassed and implements the IN.byp.1 interface.
This resource type cannot be walktested. The resource structure will contain attributes of IN.dev.1, IN.dsbl.1 and IN.byp.1.

Overview

OII Type	Supported Operations
IN.dev.1 IN.dsbl.1 IN.byp.1 IN. batteryCharger.1	Status Retrieval Enable Disable Bypass Unbypass

21.11 *Ground Fault*

The resource type IN. groundFault.1 models the ground fault input of the power supply. It provides the same interface as the resource type IN.dev.1. It can be disabled and implements the IN.dsbl.1 interface. It can be bypassed and implements the IN.byp.1 interface. This resource type cannot be walktested. The resource structure contains attributes of IN.dev.1, IN.dsbl.1 and IN.byp.1.

Overview

OII Type	Supported Operations
IN.dev.1 IN.dsbl.1 IN.byp.1 IN.groundFault.1	Status Retrieval Enable Disable Bypass Unbypass

21.12 *Power Supply CAN Outputs*

The resource type IN.psCanOp.1 models the CAN outputs on the MAP Power Supply.
It provides the same interface as the resource type IN.dev.1. This resource type cannot be disabled, bypassed or walktested. The resource structure will contain attributes of IN.dev.1.

Overview

OII Type	Supported Operations
IN.dev.1 IN.psCanOp.1	Status Retrieval

21.13 *LSNGateway*

The resource type IN.lsnGateway.1 extends the IN.dev.1 resource type with an additional firmware version command. It models the LSN Gateway. It can be disabled and implements the IN.dsbl.1 interface. This resource type cannot be bypassed or walktested. The resource structure will contain attributes of IN.dev.1 and IN.dsbl.1.

Overview

OII Type	Supported Operations
IN.dev.1 IN.dsbl.1 IN.lsnGateway.1	Status Retrieval Enable Disable Firmware Version

21.13.1 Firmware Version

The firmware version command retrieves the version of the firmware running on the LSN Gateway.

Request

Verb	*POST*
Query Parameters	(none)

Request Body

Name	Type	Value Range/Format	Description
@cmd	String	FWVERSION	

Response (Success)

Return Code	200
Content	Firmware Version Object Structure

Object Structure

Name	Type	Value Range/Format	Description
firmwareVersion	String	“1.0.1”	Firmware version of the LSN Gateway in format MAJOR.MINOR.MICRO

Response (Error)

If the panel is processing a firmware version request already when there is another request, the panel will return a 503 (Service Unavailable) response code. A client can try again at a later point of time.

21.14 *LSN Bus*

The resource type IN.lsnBus.1 models the LSN loop and LSN Stub on the LSN Gateway.
It provides the same interface as the resource type IN.dev.1. This resource type can be disabled and implements the IN.dsbl.1 interface. This resource type cannot be bypassed or walktested. The resource structure will contain attributes of IN.dev.1 and IN.dsbl.1.

Overview

OII Type	Supported Operations
IN.dev.1 IN.lsnBus.1 IN.dsbl.1	Status Retrieval Enable Disable

21.14.1 Status Retrieval

Request

Verb	*GET*
Query Parameters	(none)

Response (Success)

Return Code	200
Content	LSN Bus object structure

Response (Error)

See standard errors defined in OII Base Specification.

21.15 *LSNAux*

The resource type IN.lsnAux.1 models the LSN Aux power on the LSN Gateway. It extends IN.dev.1 resource type. This resource type can be disabled and implements the IN.dsbl.1 interface. This resource type cannot be bypassed or walktested. The resource structure will contain attributes of IN.dev.1.

Overview

OII Type	Supported Operations
IN.dev.1 IN.dsbl.1 IN.lsnAux.1	Status Retrieval Enable Disable

21.15.1 Status Retrieval

Request

Verb	*GET*
Query Parameters	(none)

Response (Success)

Return Code	200
Content	LSN Aux object structure

Response (Error)

See standard errors defined in OII Base Specification.

21.16 *System Keypad*

This models the System Keypad of the MAP5000 system. It extends IN.dev.1 resource type with additional statuses and commands specific to the keypad. It supports a firmware version command to retrieve the firmware version running on the keypad. It can be disabled and implements the IN.dsbl.1 interface. This resource type cannot be bypassed or walktested.
The resource structure will contain attributes of IN.dev.1, IN.dsbl.1 and the attributes mentioned below.

Overview

OII Type	Supported Operations
IN.dev.1 IN.dsbl.1 IN.keypad.1	Status Retrieval Enable Disable Deactivate Activate Firmware Version

Name	Type	Value Range/Format	Description
activated	Bool	True/False	Indicates whether the keypad is activated or deactivated

Name	Type	Value Range/Format	Description
userID	String	e.g. “005” , “999”	The user ID of the user that is currently logged into the keypad. In case no user is logged in, the Empty string “” will be set. Thus, a client that is subscribed for eventing for a keypad will receive a state change event including timestamp when a user logs in or off.

Note: In addition to the behaviour of a disabled device specified in section 21.3, disabling a keypad results in the keypad being locked. The LEDs do not change their state. The screen is not turned off but no user can log in. A logged in user will be logged out.

21.16.1 Deactivate

This command initiates the deactivation of a Keypad. On a deactivated keypad, the LEDs are turned off and the screen is turned off. LEDs will not change their state. The screen will not respond to user touch. A logged in user will be logged out.

Request

Verb	*POST*
Query Parameters	(none)

Request Body:

Name	Type	Value Range/Format	Description
@cmd	String	DEACTIVATE	Indicates deactivating of the device to be started

Response (Success)

Return Code	202
Content	-

Response (Error)

Return Code	Content	Description
409 (Conflict)	Keypad disabled	Response if System Keypad is disabled and hence cannot be deactivated.

Additionally see standard errors defined in OII Base Specification.

21.16.2 Activate

Request

Verb	*POST*
Query Parameters	(none)

Request Body:

Name	Type	Value Range/Format	Description
@cmd	String	ACTIVATE	Initiates activating the device

Response (Success)

Return Code	202
Content	-

Response (Error)

Return Code	Content	Description
409 (Conflict)	Keypad disabled	Response if the System Keypad is disabled.

See standard errors defined in OII Base Specification.

21.16.3 Firmware Version

The firmware version command retrieves the version of the firmware running on the System Keypad.

Request

Verb	*POST*
Query Parameters	(none)

Request Body

Name	Type	Value Range/Format	Description
@cmd	String	FWVERSION	

Response (Success)

Return Code	200
Content	Firmware Version Object Structure

Object Structure

Name	Type	Value Range/Format	Description
firmwareVersion	String	"1.0.1"	Firmware version of the System Keypad in format MAJOR.MINOR.MICRO

Response (Error)

If the panel is processing a firmware version request already when there is another request, the panel will return a 503 (Service Unavailable) response code. A client can try again at a later point of time.

21.17 *Output*

The resource type IN.output.1 extends the type IN.dev.1 with additional output state information and commands to turn on/off. It can be disabled and implements the IN.dsbl.1 interface.
This resource type cannot be bypassed or walktested. The resource structure will contain attributes of IN.dev.1 and IN.dsbl.1 in addition to the attributes mentioned below.

Overview

OII Type	Supported Operations
IN.dev.1 IN.dsbl.1 IN.output.1	Status Retrieval Enable Disable ON OFF

Name	Type	Value Range/Format	Description
on	Bool	True/False	True if output is on else false

21.17.1 ON

This operation initiates turning on the output.

Note: The pattern the output is turned on to is configured through the remote configuration tool.

Request

Verb	*POST*
Query Parameters	(none)

Request Body:

Name	Type	Value Range/Format	Description
@cmd	String	ON	Indicates initiating the turning on of the output

Response (Success)

Return Code	202
Content	-

Response (Error)

See standard errors defined in OII Base Specification.

21.17.2 OFF

This operation initiates turning off the output.

Note: Handling an incident will result in an output configured for the incident to turn off. Another occurrence of the incident will cause the output to turn on again.

Turning off an output using the OFF command directly changes the state of the output to off state. An output that is turned on because of an incident, turned off by issuing the OFF command (without the incident being silenced) will not turn-on on the occurrence of another incident of the same type.

Request

Verb	*POST*
Query Parameters	(none)

Request Body:

Name	Type	Value Range/Format	Description
@cmd	String	OFF	Indicates initiating the turning off of the output

Response (Success)

Return Code	202
Content	-

Response (Error)

See standard errors defined in OII Base Specification.

21.18 *Point*

A point is a detector such as a PIR or an EOL contact. The resource type IN.point.1 extends the IN.dev.1 type with additional property of active. This resource type can be disabled and implements the IN.dsbl.1 interface.

Points may be bypassable or walktestable depending on the configuration. An individual resource of a point will further implement the IN.byp.1 interface if bypassable and the IN.walkT.1 if it is walktestable.

The resource structure will contain attributes of IN.dev.1 and IN.dsbl.1 in addition to the attributes mentioned below. If configured to be bypassed and walktested, it will also contain attributes of IN.byp.1 and IN.walkT.1 respectively.

Overview

OII Type	Supported Operations
IN.dev.1 IN.dsbl.1 IN.point.1	Status Retrieval Enable Disable

Name	Type	Value Range/Format	Description
active	Bool	True/False	True if point is active else false

21.19 *Fire Detector*

Fire detector is an extension to the IN.point.1 type and is used to model fire detectors in the system. It adds details on the triggered sensor in the scope of a walktest. This resource type can be disabled and implements the IN.dsbl.1 interface.

Similar as a point, a fire detector may be configured to be walktestable or bypassable. Thus, the IN.walkT.1 and IN.byp.1 interface may be supported by an individual resource depending on the panel configuration. The resource structure will contain attributes of IN.dev.1, IN.dsbl.1 and IN.point.1 in addition to the attributes mentioned below. If configured to be bypassed and walktested, it will also contain attributes of IN.byp.1 and IN.walkT.1 respectively.

Overview

OII Type	Supported Operations
IN.dev.1 IN.dsbl.1 IN.point.1 IN.fireDetector.1	Status Retrieval Enable Disable

Name	Type	Value Range/Format	Description

testedSensors	Array of objects	[
		{
		type:“o”,
		tested”:true/false
		},
		{
		type:“t”,
		tested”:true/false
		},
		{
		type:“c”,
		tested”:true/false
		}
		]
			Array that contains one object for each sensor of this particular device. When device is in walktest, the array contains objects with the possible types “o”, “t” or “c” and its combinations. The tested field indicates whether the sensor has been tested successfully. The array is only filled while walktest is active. Otherwise it is empty (i.e. [])

21.20 *Keyswitch*

The IN.keyswitch.1 resource type extends the IN.dev.1 type with additional property of active. It can be disabled and implements the IN.dsbl.1 interface. Keyswitches are not bypassable and will not implement the IN.byp.1 interface. Walktest of keyswitches is currently not supported by MAP5000.

The resource structure will contain attributes of IN.dev.1 and IN.dsbl.1 in addition to the attributes mentioned below.

Overview

OII Type	Supported Operations
IN.dev.1 IN.dsbl.1 IN.keyswitch.1	Status Retrieval Enable Disable

Name	Type	Value Range/Format	Description
active	Bool	True/False	True if keyswitch is active else false

21.21 *Arming Device*

The type IN.armingDevice.1 extends the IN.dev.1 type and models the arming devices on MAP 5000. Currently this resource type doesn't represent a device on its own but will be inherited by Blocklock and smartkey (see following sections).

The resource structure will contain attributes of IN.dev.1 and IN.dsbl.1 in addition to the attributes mentioned below.

Overview

OII Type	Supported Operations
IN.dev.1 IN.dsbl.1 IN.armingDevice.1	Status Retrieval Enable Disable

Name	Type	Value Range/Format	Description
locked	Bool	True/False	True if arming device is locked (resulting in the door being locked) else false
usable	Bool	True/False	True if the arming device can be used to arm/disarm the area

21.22 *Blocklock*

The type IN.blocklock.1 is an additional resource type which provides the same interface as the IN.armingDevice.1 type and models the blocklock on MAP 5000. It can be disabled and implements the IN.dsbl.1 interface. It can be bypassed and implements the IN.byp.1 interface. This resource type cannot be walktested.

The resource structure will contain attributes of IN.dev.1, IN.dsbl.1, IN.byp.1 and IN.armingDevice.1.

Overview

OII Type	Supported Operations
IN.dev.1 IN.dsbl.1 IN.armingDevice.1 IN.blocklock.1 IN.byp.1	Status Retrieval Enable Disable Bypass Unbypass

21.23 *Smartkey*

The type IN.smartkey.1 extends the IN.armingDevice and models the smartkey. It adds additional information on update status of the smartkey. It can be disabled and implements the IN.dsbl.1 interface. It can be bypassed and implements the IN.byp.1 interface. This resource type cannot be walktested.

The resource structure will contain attributes of IN.dev.1, IN.dsbl.1, IN.byp.1 and IN.armingDevice.1 in addition to the attributes mentioned below.

Overview

OII Type	Supported Operations
IN.dev.1 IN.dsbl.1 IN.armingDevice.1 IN.smartkey.1 IN.byp.1	Status Retrieval Enable Disable Bypass Unbypass

Name	Type	Value Range/Format	Description
update	String	“NONE”, “RECOMMENDED”, “REQUIRED”, “UPDATING”	Indicates if the user data in the smartkey is up to date NONE : User data in the smartkey is up to date RECOMMENDED: User data has changed and the smartkey requires an update but everything is still working without constraints.

Name	Type	Value Range/Format	Description
			REQUIRED : User data has changed and the smartkey requires an update with the smartkey working with constraints UPDATING: User data of the smartkey is in the process of being updated.

21.24 *Coupler*

The type IN.coupler.1 models the couplers and provides the same interface as IN.dev.1 type.
It can be disabled and implements the IN.dsbl.1 interface. It cannot be bypassed or walktested.

The resource structure will contain attributes of IN.dev.1, IN.dsbl.1.

Overview

OII Type	Supported Operations
IN.dev.1 IN.dsbl.1 IN.coupler.1	Status Retrieval Enable Disable

21.25 *Communicator*

The type IN.communicator.1 models the communicator (AT2000 and AT3000) and provides the same interface as IN.dev.1 type. It can be disabled and implements the IN.dsbl.1 interface.
It cannot be bypassed or walktested.

The resource structure will contain attributes of IN.dev.1, IN.dsbl.1.

Overview

OII Type	Supported Operations
IN.dev.1 IN.dsbl.1 IN.communicator.1	Status Retrieval Enable Disable

21.26 *Printer*

The resource IN.printer.1 extends the IN.dev.1 type with additional parameters such as cover open and paper low. It models the printer device. It can be disabled and implements the IN.dsbl.1 interface. It cannot be bypassed or walktested.

The resource structure will contain attributes of IN.dev.1, IN.dsbl.1 in addition to the attributes mentioned below.

Overview

OII Type	Supported Operations
IN.dev.1 IN.dsbl.1 IN.printer.1	Status Retrieval Enable Disable

Name	Type	Value Range/Format	Description
coverOpen	Bool	True/False	True if printer cover is open, otherwise false
paperLow	Bool	True/False	True if printer is low on paper, else false

21.27 *Power Device*

The type IN.powerdevice.1 models power devices configured on couplers on the LSN. It provides the same interface as IN.dev.1 type. It can be disabled and implements the IN.dsbl.1 interface. It cannot be bypassed or walktested.

The resource structure will contain attributes of IN.dev.1, IN.dsbl.1.

Overview

OII Type	Supported Operations
IN.dev.1 IN.dsbl.1 IN.powerDevice.1	Status Retrieval Enable Disable

22 Supervised Connection

The resource type IN. supervConn.1 is a basic type used to model all supervised connections or communication paths in MAP. This resource provides status of the supervised connections. These resources can be disabled and when disabled the panel wouldn't supervise the connections.

Overview

OII Type	Supported Operations
IN. supervConn.1 IN.dsbl.1	Status Retrieval Enable Disable

Object structure

Name	Type	Value Range/Format	Description
@type	Array of strings	["IN. supervConn.1"]	Fixed type identifier
@self	String	URL starting with "/"	Link to the current resource
incs	Array	Array of links to currently pending incidents related to this device	A list of incidents that relate to the supervised connection.
enabled	Bool	True/False	True if detection of incidents from the resource has been disabled else true

22.1 *Status Retrieval*

Request

Verb	*GET*
Query Parameters	(none)

Response (Success)

Return Code	200
Content	Supervised connection structure

Response (Error)

See standard errors defined in OII Base Specification.

22.2 *BIS Supervision*

The resource IN.supervBis.1 models the supervision of the BIS connection to MAP. It provides the same interface as the IN.supervConn.1 resource type. It cannot be bypassed or walktested. The resource structure will contain attributes of IN. supervConn.1 and IN.dsbl.1.

Overview

OII Type	Supported Operations
IN. supervConn.1 IN.dsbl.1 IN. supervBis.1	Status Retrieval Enable Disable

22.3 *IPC Supervision*

The resource type IN.supervIpc.1 models the supervision of the IP Communicator communication paths. It extends the IN.supervConn.1 with the path information. This resource type cannot be bypassed or walktested. The resource structure will contain attributes of IN. supervConn.1 and IN.dsbl.1.

Overview

OII Type	Supported Operations
IN. supervConn.1 IN.dsbl.1 IN. supervIpc.1	Status Retrieval Enable Disable

Name	Type	Value Range/Format	Description
path	String	“ethernet”, “wireless”	Indicates the IPC communication path

22.4 *OII Supervision*

The resource type IN.oiiSupvs.1 models the supervision of the OII users over the interface. It extends the IN.supervConn.1 with the user that is being supervised. This resource type cannot be bypassed or walktested. The resource structure will contain attributes of IN. supervConn.1 and IN.dsbl.1 in addition to the attributes mentioned below.

Overview

OII Type	Supported Operations
IN. supervConn.1 IN.dsbl.1 IN. oiiSupvs.1	Status Retrieval Enable Disable

Name	Type	Value Range/Format	Description
user	String	3 digits ranging from 004 to 999	The OII user id that is being supervised

22.5 *OII*

The resource IN.oii.1 models the Open Intrusion Interface of the MAP. It provides information of whether there was a brute force login attempt. The resource can be disabled to prevent MAP from detecting incidents related to it. It can be disabled and implements the IN.dsbl.1 interface.
This resource type cannot be bypassed or walktested.

Overview

OII Type	Supported Operations
IN. oii.1 IN.dsbl.1	Status Retrieval Enable Disable

Object structure

Name	Type	Value Range/Format	Description
@type	Array of strings	[“IN. oii.1”]	Fixed type identifier
@self	String	URL starting with “/”	Link to the current resource
enabled	Bool	True/False	True if detection of incidents from the resource has been disabled else true

Name	Type	Value Range/Format	Description
incs	Array	Array of links to currently pending incidents related to this resource	A list of incidents that relate to open intrusion interface.

22.5.1 Status Retrieval

Request

Verb	*GET*
Query Parameters	(none)

Response (Success)

Return Code	200
Content	OII resource structure

Response (Error)

See standard errors defined in OII Base Specification.

23 Reactionless Interface

The MAP can be configured to provide the OII as a “reactionless” interface. This means that data from the MAP will be provided, but all operations on the MAP (e.g. arming, bypassing, etc.) will not be allowed. In cases when the reactionless option is activated in the panel, a client will receive a 403 Forbidden error, whenever it tries to execute a command on the MAP.

24 List of changes across OII Resource Model versions

24.1 *OII Resource Model Version 1.0 and 1.1*

24.1.1 Panel Resource

The following attributes added to the Panel resource.

isPanelLoaded	Bool	True/False
restartImminent	Bool	True/False

The panel resource type modified to **IN.panel.1.1**

24.1.2 Incident Resource

The following attribute added to the Incident resource.

extInc	Bool	True/False

The incident resource type modified to **IN.inc.1.1**

25 Change History

Date	Description	Author
09.05.2014	First version for external review	Barisic, Moorthi
17.06.2014	Updated to align with current SW status	
27.08.2014	Modified Network Settings – need for permission and reboot – Section 14 Modified History Status retrieval – Section 8	Arun Ram Moorthi
09.08.2014	Updated section 11 – Arming Info	Arun Ram Moorthi
06.10.2014	Updated section 2 for internal programs	Arun Ram Moorthi
21.10.2014	Updated section 9 – History for IPC history retrieval	Arun Ram Moorthi
13.11.2014	Updated to new naming guidelines (camelcase, @type, @self, no namespaces in property keys) Added @cmd properties to distinguish commands Removed reason persists in incidents	Daniel Barisic
21.11.2014	Resource urls – made consistent to be in lower case	Arun Ram Moorthi
10.12.2014	Added section on Devices	Daniel Barisic Arun Ram Moorthi
11.12.2014	Added Panel resource Updated User and User List resources Updated devices post discussion with Daniel Added list resources for devices Added description for subscription to all elements in the system Changed arming info to command instead of being a resource	Arun Ram Moorthi Daniel Barisic
12.12.2014	Changed date format for history Updated examples	Daniel Barisic

Date	Description	Author
14.12.2014	Added application error codes for requests Added configuration description	Arun Ram Daniel Barisic
15.12.2014	Added description for motion detector tests Add description under devices section Added additional application error codes Removed Alarm System Over Current in incident categorization Added description for device list resources under Devices section Added the IN.dsbl.1 interface Updated start walktest on area list resource and activate in internal program list resources Version update to 1.0 (removed draft)	Arun Ram
09.01.2015	Section 6 : Panel : Object Structure : corrected case of “failsafe” attribute Section 8 : User : Object Structure : corrected case for “active”, ”type”, ”name”, ”language” attributes Section 8.2 : - Activate: Removed installer:Standard and installer:OneTime as the type of users that the command can be used on. - Removed error code “Date not accepted in ACTIVATE command” - Removed error code Invalid command for temporary type of users. Use “ACTIVATEUNTIL” Section 8.3 : - Activate Until: Removed installer:temporary as the type of user that the command can’t be used on. - Removed error codes for “Date not specified”, “Date not within expected range”,” Date not in expected format”, “Invalid command for standard and one time type of users. Use “ACTIVATE”” Section 8.4 : - Deactivate : added user types the command can be used on - Added response error code	Arun Ram

Date	Description	Author
	Section 9 : User List Resource : corrected case for “list” attributes Section 9.2 : - Activate : Removed error codes “Date not accepted in ACTIVATE command” and “Invalid command for temporary type of users. Use “ACTIVATEUNTIL”” Section 9.3 : - Activate Until : Removed response error codes “Date not specified” , “Date not within expected range”, “Date not in expected format”, “Invalid command for standard and one time type of users. Use “ACTIVATE”” Section 10.1.2 : Incident Resource : Handle : Added response error code “Area armed” Section 15.2: Area Resource : Arm : - Modified error codes “Area not ready to arm” , “Cannot force arm area” , “Area not armable” - Removed error code for “Attempt to force arm when ready” - Added error code for “RPS connected to Panel” Section 15.3 : Disarm - Modified response error codes for “Area not ready to disarm” , “Area not armable” Section 15.4 : Start Walktest - Response error codes modified for “Walktest Already Running” , “No points to walktest” , “Area armed” Section 15.8 : Start Chime Mode - Added response error code for “Not supported by Control Panel Area” Section 15.9 : Stop Chime Mode - Added response error code for “Not supported by Control Panel Area” Section 15.10 : Start Bell Test - Modified Response code for Start Bell Test command	

Date	Description	Author
	Section 16.4 : Start Walktest - Modified response code for walktest on area list Section 19.2 “: Acitvate Internal Program - Modified response code for activating internal program Section 21.11.1 : System Keypad : Deactivate - Modified response code for “Keypad disabled” Section 21.11.2 : System Keypad : Activate - Modified response code for “Keypad disabled” Section 22.2 : Supervised Connection : Disable - Modified response code for “Area Armed” Section 22.3 : Supervised Connection : Enable - Modified response code for “Area Armed” Section 22.7.2 : OII : Disable - Modified response code for “Area Armed” Section 22.7.2 : OII : Enable - Modified response code for “Area Armed” Section 22.8.1 : Disable : Disable - Modified response code for “Area Armed” Section 22.8.2 : Disable : Enable : - Modified response code for “Area Armed” Section 22.9.1 : Bypass: Bypass: - Modified response code for “Area Armed” - Added comment on configurability of bypass in an armed area Section 22.9.2 : Bypass: Unbypass: - Modified response code for “Area Armed”	

Date	Description	Author
	- Added comment on configurability of unbypass in an armed area	
18.01.2015	Section 15.11 : ArmingInfo object structure - Added attribute "tooManyBypassedDevices" - Response code corrected to 200 (from 204)	Arun Ram
17.03.2015	Removed all references to allowed flag. Modified success response to POST commands from 204 ("OK") to 202 ("Accepted") (except for POST commands on User and User list resources) Modified success and error response codes for POST on list resources with and without atomic set. Section 2 : Resource Model Overview : - Updated resources list and supported commands Section 3: Description : - Updated main resources list Section 5 : Configuration : - Added Internal Program configuration - Added POINT.MCP, POINT.VIRTUAL, KEYSWITCH.VIRTUALSTATIC, KEYSWITCH.VIRTUALDYNAMIC, PSCANOUTPUT device types in device configuration Section 6: - Changed values for cfgStatus Section 8: - Changed error code for ACTIVATE on endUser:Temporary from 400 to 403 Section 10: Incident : - Updated incident handling scenarios and figures - Added Trouble.System.Initialization and Trouble.System.Failure incidents in incident categorization Section 13 : History - Added error response code 503	Daniel Barisic Arun Ram

Date	Description	Author
	Section 15.2 : Arm - Removed 409 (Conflict) “Cannot Force Arm error” - Change “not Armable” to “Not user armable”. Same for disarm Section 15.11 : Arming Info - Added error response code 503 Section 16 : Area List - Added Start/Stop chime mode as supported operation - Walktest Sucess Response : Added comment Section 19 : Internal Program : - Added IP Arming Info in supported operations - Modified error response content for Activate - Added IP Arming Info command description Section 20 : Internal Program List : - Modified error response content for Activate Section 21 : Devices : - Added section 21.2 : Device Lists - Section 21.3 – Added 409 (“Conflict”) “Area not configured for device” response for disable and enable commands. - Section 21.4 - Added 409 (“Conflict”) “Area not configured for device” response for bypass and unbypass commands. - Section 21.6 : DeModule : Added Firmware Version command - Section 21.7 : Power supply : Added Firmware Version command - Added Section 21.12 : Power Supply CAN Outputs - Section 21.13 : LSN Gateway : Added Firmware Version command	

Date	Description	Author
	- Section 21.16: System Keypad : Added Firmware Version command. - Removed Bosch Data Bus device - Section 21.19 : Fire Detectors : Modified testedSensors in object structure	
07.04.2015	Section 4 Event Notification - Changed “/users/*” to “/user/*”	Daniel Barisic
22.04.2015	Section 7 Time - Specified timestamp range and format for set time Section 8 User - Adjusted timestamp range to 1970 – 2036 Section 12 Network Settings - Added section 12.1 “Status Retrieval” Section 17 Walktest - Added fields for user, interface and wtStatus to walktest resource Section 21.4 Bypass - Added 409 Conflict for bypass on a device that is in walktest	Daniel Barisic
28.04.2015	Section 21.16 System Keypad - extended note describing behaviour of a disabled keypad Section 21.5.1 Diagnose: - Added time zone information as a part of lastWalktestDate attribute Section 4 Event Notification - Added subscription details for walktests Section 21.23 Smartkey - Added update status of “UPDATING”	Arun Ram Moorthi
29.05.2015	Section 21.19 Fire Detector - Corrected typos Section 9.2 User List : Activate Section 9.3 User List : Activate Until - Corrected typos	Arun Ram Moorthi
06.08.2015	Minor editorial changes	Barisic

Date	Description	Author
05.11.2015	Section 2 : Updated OII Resource Model Version to 1.1 Updated Section 15 : Area : Object Structure - oiiArmable to indicate that area containing smartkeys will now be oiiArmable Section 6 : Panel Resource : Object Structure - Added isPanelLoaded and restartImminent Added Section 24 : - List of changes across OII Resource Model versions	Arun Ram Moorthi
23.03.2016	Section 10 : Incident Resource : - Added attribute ‘extInc’. - Updated incident resource type to IN.inc.1.1	Arun Ram Moorthi
11.04.2019	Section 8 Users: - Added 2 new languages, RO and LT Section 10.2.2.1 Alarms > Intrusion: - Added new sub-category ‘Amok’	Heinrich Erhard

26 References

[HTTP]	http://www.ietf.org/rfc/rfc2616.txt
[URL]	http://www.ietf.org/rfc/rfc3986.txt. HTTP spec. references http://www.ietf.org/rfc/rfc2396.txt but RFC3986 obsoletes RFC2396.
[JSON]	Specification http://www.ecma-international.org/publications/files/ECMA-ST/ECMA-404.pdf, Official Website http://www.json.org/

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